Chapter 3
Ethical Principles for Trustworthy AI

3.1 Overview of the European AI Strategy

Policymakers across the world are increasingly focusing on the question of how AI should be regulated in order to tackle the risks associated with the development of AI-enabled technologies and their impact on society. It is reminded that, in January 2017, the European Parliament's Legal Affairs (JURI) Committee took the initiative to draft a report tabling a motion for a Resolution of a legislative character and including recommendations to the Commission on the civil-law and ethical aspects of robotics.[1] The European Parliament adopted, on 16 February 2017, a Resolution containing recommendations to the Commission on Civil Law Rules on Robotics.[2] The Resolution made wide ranging recommendations and asked the Commission to assess the impact of AI and present a legislative proposal laying down a set of civil law rules on robotics and artificial intelligence. The Resolution called for EU legislation introducing a register of robots, setting up an EU Agency for Robotics and AI (tasked with providing the technical, ethical and regulatory expertise needed in an AI-driven environment) and laying down principles of civil liability for damages caused by robots. Such legislation should be complemented by ethical codes of conduct.

In particular, the Resolution proposed, as an annex, two draft codes of conduct—a Code of Ethical Conduct for Robotics Engineers and a Code for Research Ethics Committees. The first code, which could update and complement the existing EU legal framework, put forward four ethical principles in robotics engineering:

[1] Report of the Committee on Legal Affairs of the European Parliament with recommendations to the Commission on Civil Law Rules on Robotics—Motion for a European Parliament Resolution (2015/2103(INL)), 27.1.2017, A8-0005/2017.

[2] European Parliament Resolution of 16 February 2017 with recommendations to the Commission on Civil Law Rules on Robotics (2015/2103(INL)), (2018/C 252/25), OJ C 252, 18.7.2018, pp. 239–257 (hereafter: "European Parliament Resolution of 16 February 2017").

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(i) beneficence (robots should act in the best interests of humans); (ii) non-maleficence (robots should not harm humans); (iii) autonomy (human interaction with robots should be voluntary—the capacity to make an informed, un-coerced decision about the terms of interaction with robots;); (iv) justice (the benefits of robotics should be distributed fairly—also, affordability of homecare and healthcare robots in particular).[3] AI and robots increase their interaction with humans in very diverse fields and the risks posed by these new interactions should be properly tackled, ensuring that a set of fundamental values is translated into every stage of contact between robots, AI and humans. In this process, the aim of the ethical conduct code is to ensure that robots and AI operate in accordance with legal safety and ethical standards. Special emphasis should be given to human safety, privacy, integrity, dignity and autonomy.[4] The European Parliament considered it essential to guarantee that humans have control over intelligent machines at all times,[5] that robotics engineers and designers should be accountable for the social, environmental and human health impacts that robotics may impose (accountability),[6] that they should respect human well-being, health and rights, guarantee

[3]European Parliament Resolution of 16 February 2017, Annex to the Resolution: Recommendations as to the content of the proposal requested, p. 253. See also points 13 and 14 of the Resolution: "(13) . . . the guiding ethical framework should be based on the principles of beneficence, non-maleficence, autonomy and justice, on the principles and values enshrined in Article 2 of the Treaty on European Union and in the Charter of Fundamental Rights, such as human dignity, equality, justice and equity, non-discrimination, informed consent, private and family life and data protection, as well as on other underlying principles and values of the Union law, such as non-stigmatisation, transparency, autonomy, individual responsibility and social responsibility, and on existing ethical practices and codes"; "(14) . . . special attention should be paid to robots that represent a significant threat to confidentiality owing to their placement in traditionally protected and private spheres and because they are able to extract and send personal and sensitive data".

[4]European Parliament Resolution of 16 February 2017, point 10, p. 244: "the potential for empowerment through the use of robotics is nuanced by a set of tensions or risks and should be seriously assessed from the point of view of human safety, health and security; freedom, privacy, integrity and dignity; self-determination and non-discrimination, and personal data protection". See also: recital O; point 13; Annex, at p. 253 and p. 254: "Robotics research activities should respect fundamental rights and be conducted in the interests of the well-being and self-determination of the individual and society at large in their design, implementation, dissemination and use. Human dignity and autonomy – both physical and psychological – is always to be respected".

[5]Ibid., point 3, at p. 243.

[6]European Parliament Resolution of 16 February 2017, Annex to the Resolution: Recommendations as to the content of the proposal requested, at pp. 244 and 255. See also recital M (p. 240): "the trend towards automation requires that those involved in the development and commercialisation of AI applications build in security and ethics at the outset, thereby recognizing that they must be prepared to accept legal liability for the quality of the technology they produce".

transparency and inclusiveness,[7] and disclose promptly factors that might endanger the public or the environment (safety).[8]

Against this background, i.e. the European Parliament's Resolution of 16 February 2017 containing recommendations to the Commission on Civil Law Rules on Robotics, and in response to the European Council's call[9] to put forward a European approach to AI, the Commission published, on 25 April 2018, its Communication on "Artificial Intelligence for Europe".[10] This "AI for Europe" Communication presented the European AI strategy, recognising that AI can benefit the whole of society and the economy but, at the same time, also brings with it new challenges and raises legal and ethical questions. This EU strategy placed people at the centre of the development of AI, i.e. human-centric AI, and proposed a three-pronged approach, namely to boost the EU's technological and industrial capacity and AI uptake across the economy, prepare for socio-economic changes, and ensure an appropriate ethical and legal framework. The "AI for Europe" Communication laid the groundwork for the 'Coordinated Plan on AI' and outlined the steps required to achieve a more committed alignment of resources and goals between Member States in order to take advantage of the opportunities offered by AI and to address the new challenges that it brings.

The proposal for a coordinated plan on AI was endorsed by the European Council[11] which, at its meeting on 28 June 2018, invited the Commission to work with Member States on a coordinated plan on Artificial Intelligence, building on its Communication "Artificial Intelligence for Europe", to create synergies, pool data and increase joint investments. As a result, and in order to coordinate the EU Member States' national AI strategies, the Commission submitted on 7 December 2018 the Communication "Coordinated Plan on Artificial Intelligence",[12]

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[7] "Inclusiveness allows for participation in decision-making processes by all stakeholders involved in or concerned by robotics research activities". European Parliament Resolution of 16 February 2017, Annex to the Resolution: Recommendations as to the content of the proposal requested, p. 254.

[8] "Robot designers should consider and respect people's physical wellbeing, safety, health and rights. A robotics engineer must preserve human wellbeing, while also respecting human rights, and disclose promptly factors that might endanger the public or the environment"; "Precaution: Robotics research activities should be conducted in accordance with the precautionary principle, anticipating potential safety impacts of outcomes and taking due precautions, proportional to the level of protection, while encouraging progress for the benefit of society and the environment". European Parliament Resolution of 16 February 2017, Annex to the Resolution: Recommendations as to the content of the proposal requested, p. 254.

[9] General Secretariat of the Council, Subject: European Council meeting (19 October 2017) – Conclusions, Brussels, 19 October 2017 (OR. en), EUCO 14/17, at Section II – Digital Europe.

[10] Commission Communication "Artificial Intelligence for Europe", COM(2018) 237 final, Brussels, 25.4.2018.

[11] European Council conclusions, 28 June 2018: https://www.consilium.europa.eu/en/press/press-releases/2018/06/29/20180628-euco-conclusions-final/.

[12] Commission Communication "Coordinated Plan on Artificial Intelligence", COM(2018) 795 final, Brussels, 7.12.2018.

accompanied by its Annex entitled "Coordinated Plan on the Development and Use of Artificial Intelligence Made in Europe - 2018"[13] prepared by the Member States, Norway, Switzerland and the Commission in the context of the work of the Member States' Group on Digitising European Industry and Artificial Intelligence.

3.2 Building Trust in Human-Centric Artificial Intelligence

Building on the above-mentioned groundwork, following the call from the European Parliament to update and complement the existing EU legal framework with guiding ethical principles, and based on the EU's 'human-centric' approach to AI that is respectful of European values and principles, the Commission's High-Level Expert Group on AI (comprising 52 independent experts representing academia, industry, and civil society, who act as an advisory group to the Commission on ethics, investment and policymaking and work on outlining a long-term AI strategy) published a first draft of the ethics guidelines on 18 December 2018.[14]

Following further deliberations by the High-Level Expert Group (AI HLEG) in light of discussions on the European AI Alliance, a stakeholder open consultation (which generated feedback from more than 500 contributors) and meetings with representatives from Member States to gather feedback, the Ethics Guidelines for trustworthy AI were revised and published in April 2019,[15] putting forward trust as a prerequisite to ensure a human-centric approach to AI and aiming at offering guidance on how to foster and secure the development of ethical AI systems in the EU. The AI HLEG also prepared a revised document which elaborates on a definition of Artificial Intelligence used for the purpose of its deliverables.[16]

The AI HLEG's Ethics Guidelines for Trustworthy AI were one of the clearest demonstrations of the EU's ambition to become a leader in 'ethical AI'. Although they solely express the opinion of the AI HLEG and may not be regarded as reflecting an official position of the Commission, the ethics guidelines nonetheless constitute an insight into the direction that Europe is heading in, namely towards 'human-centric and trustworthy AI'. The guidelines contain a variety of chapters ranging from AI's compliance with fundamental rights, technical and non-technical

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[13] Annex to the Commission Communication 'Coordinated Plan on Artificial Intelligence', COM (2018) 795 final, Brussels, 7.12.2018 - Coordinated Plan on the Development and Use of Artificial Intelligence Made in Europe – 2018.

[14] "Draft Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 18 December 2018.

[15] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019 (hereafter: "Ethics Guidelines for Trustworthy AI, 2019").

[16] "A definition of AI: Main capabilities and scientific disciplines – Definition developed for the purpose of the deliverables of the High-Level Expert Group on AI", High-Level Expert Group on Artificial Intelligence, 18 December 2018.

methods to implement the outlined ‘trustworthy AI’ to an assessment list and case studies for ‘trustworthy AI’.

The overall aim of the ethics guidelines is—apart from establishing an ethical level playing field across all Member States, which should be implemented by developers, suppliers and users of AI within the EU—to bring a European ethical approach to the global stage, i.e. to stimulate discussion of ethical frameworks for AI “at a global level” and, therefore, to build an international consensus on AI ethics guidelines. As a result, Europe could become a leader through this approach of using ethics as “inspiration to develop a unique brand of AI” (“AI made in Europe”) and thus gain a competitive advantage for the European industry.

Furthermore, the ethics guidelines served as starting point for discussion on “Trustworthy AI made in Europe”. With a view to this objective, the Commission published on 8 April 2020 its Communication “Building Trust in Human-Centric Artificial Intelligence”,[17] where it stressed that AI’s trustworthiness must be ensured since people will only be able to confidently reap the benefits of a technology that they can fully trust: “building on this existing regulatory framework, there is a need for ethics guidelines for the development and use of AI due to its novelty and the specific challenges this technology brings. Only if AI is developed and used in a way that respects widely-shared ethical values, it can be considered trustworthy”.[18] The Commission welcomed the ethics guidelines (“the Commission welcomes the work of the AI high-level expert group and considers it valuable input for its policy-making”),[19] it mentioned that the work of the High-Level Expert Group on AI was positively received by Member States and stakeholders[20] and, in an attempt to create an environment of trust for the successful development and use of AI, (although recognising that the guidelines are non-binding and as such do not create any new legal obligations) it encouraged all stakeholders to implement the document’s key requirements.[21]

[17] Commission Communication, “Building Trust in Human-Centric Artificial Intelligence”, COM (2019) 168 final, Brussels, 8.4.2019.

[18] Ibid., at p. 9.

[19] Ibid., at pp. 4 and 9.

[20] Ibid., at p. 3. “In their feedback so far, stakeholders overall have welcomed the practical nature of the guidelines and the concrete guidance they offer to developers, suppliers and users of AI on how to ensure trustworthiness”.

[21] Ibid., at pp. 4 and 9: “The Commission supports the following key requirements for trustworthy AI, which are based on European values. It encourages stakeholders to apply the requirements and to test the assessment list that operationalises them in order to create the right environment of trust for the successful development and use of AI” (at p. 4); “based on the key requirements for AI to be considered trustworthy, the Commission will now launch a targeted piloting phase to ensure that the resulting ethical guidelines for AI development and use can be implemented in practice” (at p. 9).

3.3 Notion of Human-Centric and Trustworthy AI

The core principle of the ethics guidelines is that the EU must develop a 'human-centric' approach to AI that is respectful of European values and principles. In particular, according to the guidelines, the Union's values and fundamental rights should always be at the forefront of the AI landscape. AI should serve people, aiming at increasing human well-being and benefiting people and society as a whole. The power of AI should be placed at the service of human progress, human dignity, freedom, pluralism, justice, and equality. Trust is a prerequisite to ensure a human-centric approach to AI—i.e. an approach that places people at the centre of the development of AI—and this is why the regulatory framework for AI should be based on the Union's values and in line with the EU Charter of Fundamental Rights.

As the High-Level Expert Group on AI pointed out: "The human-centric approach to AI strives to ensure that human values are central to the way in which AI systems are developed, deployed, used and monitored, by ensuring respect for fundamental rights, including those set out in the Treaties of the European Union and Charter of Fundamental Rights of the European Union, all of which are united by reference to a common foundation rooted in respect for human dignity, in which the human being enjoys a unique and inalienable moral status. This also entails consideration of the natural environment and of other living beings that are part of the human ecosystem, as well as a sustainable approach enabling the flourishing of future generations to come".[22] "We believe that AI has the potential to significantly transform society. AI is not an end in itself, but rather a promising means to increase human flourishing, thereby enhancing individual and societal well-being and the common good, as well as bringing progress and innovation. In particular, AI systems can help to facilitate the achievement of the UN's Sustainable Development Goals, such as promoting gender balance and tackling climate change, rationalising our use of natural resources, enhancing our health, mobility and production processes, and supporting how we monitor progress against sustainability and social cohesion indicators. To do this, AI systems need to be human-centric, resting on a commitment to their use in the service of humanity and the common good, with the goal of improving human welfare and freedom. While offering great opportunities, AI systems also give rise to certain risks that must be handled appropriately and proportionately. We now have an important window of opportunity to shape their development. We want to ensure that we can trust the sociotechnical environments in which they are embedded. We also want producers of AI systems to get a competitive advantage by embedding Trustworthy AI in their products and services. This entails seeking to maximise the benefits of AI systems while at the same time preventing and minimising their risks".[23]

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[22] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019. See the definition provided in the glossary section of the Ethics Guidelines, at p. 37.
[23] Ibid., p. 4.

Also, the High-Level Expert Group on AI pointed out that the EU is based on a constitutional commitment to protect the fundamental rights of human beings, to ensure respect for the rule of law, to foster democratic freedom and promote the common good. These specific rights are reflected in Articles 2 and 3 of the Treaty of the European Union, and in the Charter of Fundamental Rights of the EU: "We believe in an approach to AI ethics based on the fundamental rights enshrined in the EU Treaties, the EU Charter and international human rights law. Respect for fundamental rights, within a framework of democracy and the rule of law, provides the most promising foundations for identifying abstract ethical principles and values, which can be operationalised in the context of AI. The EU Treaties and the EU Charter prescribe a series of fundamental rights that EU member states and EU institutions are legally obliged to respect when implementing EU law. These rights are described in the EU Charter by reference to dignity, freedoms, equality and solidarity, citizens' rights and justice. The common foundation that unites these rights can be understood as rooted in respect for human dignity – thereby reflecting what we describe as a "human-centric approach" in which the human being enjoys a unique and inalienable moral status of primacy in the civil, political, economic and social fields".[24]

As the Commission underlined, "trust is a prerequisite to ensure a human-centric approach to AI: AI is not an end in itself, but a tool that has to serve people with the ultimate aim of increasing human well-being. To achieve this, the trustworthiness of AI should be ensured. The values on which our societies are based need to be fully integrated in the way AI develops. The Union is founded on the values of respect for human dignity, freedom, democracy, equality, the rule of law and respect for human rights, including the rights of persons belonging to minorities. These values are common to the societies of all Member States in which pluralism, non-discrimination, tolerance, justice, solidarity and equality prevail. In addition, the EU Charter of Fundamental Rights brings together – in a single text – the personal, civic, political, economic and social rights enjoyed by people within the EU";[25] "AI technology should be developed in a way that puts people at its centre and is thus worthy of the public's trust. This implies that AI applications should not only be consistent with the law, but also adhere to ethical principles and ensure that their implementations avoid unintended harm. Diversity in terms of gender, racial or ethnic origin, religion or belief, disability and age should be ensured at every stage of AI development. AI applications should empower citizens and respect their fundamental rights. They should aim to enhance people's abilities, not replace them, and also enable access by people with disabilities";[26] "only if AI is developed and used in a way that respects widely-shared ethical values, it can be considered trustworthy. . . . The ethical dimension of AI is not a luxury feature or an add-on: it needs to be an

[24] Ibid., at pp. 9 and 10.

[25] Commission Communication, "Building Trust in Human-Centric Artificial Intelligence", COM (2019) 168 final, Brussels, 8.4.2019, at pp. 1 and 2.

[26] Ibid., at p. 2.

integral part of AI development. By striving towards human-centric AI based on trust, we safeguard the respect for our core societal values and carve out a distinctive trademark for Europe and its industry as a leader in cutting-edge AI that can be trusted throughout the world".[27]

3.4 A Framework for Trustworthy AI—Three Components: Lawful, Ethical, Robust AI

AI can benefit the whole of society and the economy. Nevertheless, AI also brings with it new and substantial challenges because it enables machines to "learn" and to implement decisions without human intervention. It is a technology that is now being developed and used at a rapid pace across the world and, in the near future, this kind of functionality could become standard in many types of goods and services, from smart phones to automated cars, robots and online applications. Yet, decisions taken by algorithms could result from data that is biased, mistaken, incomplete, inadequate or damaged and manipulated by cyber-attackers. Therefore, "unreflectively applying the technology as it develops would lead to problematic outcomes as well as reluctance by citizens to accept or use it. Instead, AI technology should be developed in a way that puts people at its centre and is thus worthy of the public's trust. This implies that AI applications should not only be consistent with the law, but also adhere to ethical principles and ensure that their implementations avoid unintended harm. Diversity in terms of gender, racial or ethnic origin, religion or belief, disability and age should be ensured at every stage of AI development. AI applications should empower citizens and respect their fundamental rights. They should aim to enhance people's abilities, not replace them, and also enable access by people with disabilities. Therefore, there is a need for ethics guidelines that build on the existing regulatory framework and that should be applied by developers, suppliers and users of AI in the internal market, establishing an ethical level playing field across all Member States".[28]

It is clear, therefore, that EU policy-makers have chosen to remain faithful to the EU's cultural preferences and higher standard of protection against the risks posed by AI, building on the existing regulatory framework and ensuring that European values are at the heart of creating the right environment of trust for the successful development and use of AI—unlike other more lax jurisdictions.

To that end, "Trustworthy AI" is described by the ethics guidelines as made up of three components, which should be met throughout the system's entire life cycle, namely: (1) it should be lawful, i.e. complying with all applicable laws and regulations; (2) it should be ethical, i.e. ensuring adherence to ethical principles and values. Indeed, laws are not always up to speed with technological developments or, at

[27] Ibid., at p. 9.

[28] Ibid., at p. 2.

times, they can be out of step with ethical norms. Therefore, for AI systems to be trustworthy, they should also be ethical, ensuring alignment with ethical values and norms; and (3) it should be robust, both from a technical and social perspective. Indeed, even if an ethical purpose is ensured, society must also be confident that AI systems perform in a safe, secure and reliable manner, and that specific safeguards were foreseen to prevent any unintentional harm.[29] According to the ethics guidelines, each of the aforementioned three components is necessary but not sufficient for the achievement of Trustworthy AI; all three components work cumulatively in harmony, overlap in their operation, are closely intertwined and complement each other.[30]

With regard to lawful AI in particular, the Guidelines noted the importance of having a predictable legal environment. Legal certainty is indispensable in order to ensure trust, accountability and facilitate innovation. It was stressed that AI systems cannot operate in a lawless world and that the EU has a strong and balanced regulatory framework, with high standards in terms of safety, product liability, network/systems security and protection of personal data. A number of legally binding rules already apply or are relevant to the development and use of AI systems. These legal sources include EU primary law (i.e. the Treaties of the European Union and its Charter of Fundamental Rights), EU secondary law (such as the General Data Protection Regulation, the Product Liability Directive, the Regulation on the Free Flow of Non-Personal Data, the Cybersecurity Act, anti-discrimination Directives, consumer law and Safety and Health at Work Directives), the UN Human Rights Treaties and the Council of Europe conventions (such as the European Convention on Human Rights), and various EU Member State laws. In addition, numerous domain-specific rules exist that apply to particular AI-enabled systems (such as the Medical Device Regulation in the healthcare sector).[31]

The ethics guidelines clarified that they do not discuss the legal obligations that apply to AI systems in further detail, i.e. they do not explicitly deal with the first component of Trustworthy AI (lawful AI),[32] but focus on the latter two components—that AI systems should be "ethical" and "robust". This is why the guidelines were restricted to the following self-evident statement: "we however

[29] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at pp. 2 and 5–7.

[30] In practice, however, there may be tensions between these elements (e.g. at times the scope and content of existing law might be out of step with ethical norms). It is our individual and collective responsibility as a society to work towards ensuring that all three components help to secure Trustworthy AI. This also means that policy-makers may need to review the adequacy of existing laws where these might be out of step with ethical principles. See Ethics Guidelines for Trustworthy AI, 2019, at p. 5.

[31] Ibid., at p. 6.

[32] Although the Guidelines set out a framework for achieving Trustworthy AI, they do not explicitly deal with Trustworthy AI's first component (i.e. lawful AI). Instead, the main aim of the Guidelines is to offer guidance on the second and third components of trustworthy AI, i.e. fostering and securing ethical and robust AI. See Ethics Guidelines for Trustworthy AI, 2019, at pp. 2 and 6.

recall that it is the duty of any natural or legal person to comply with laws – whether applicable today or adopted in the future according to the development of AI. These Guidelines proceed on the assumption that all legal rights and obligations that apply to the processes and activities involved in developing, deploying and using AI systems remain mandatory and must be duly observed".[33]

3.5 Foundations of Trustworthy AI: Fundamental Rights and Ethical Principles for Ethical and Robust AI

The HLEG AI ethics guidelines set out the foundations of Trustworthy AI by laying out a fundamental rights-based approach. This foundation is grounded in fundamental rights and reflected by four ethical principles that must be adhered to in order to ensure ethical and robust AI. As already mentioned, ethical and robust AI are complementary, and the principles put forward in the ethics guidelines as well as the requirements derived from these principles address both components.

Ethical AI, i.e. ensuring alignment with ethical values and norms, can serve multiple purposes, including the protection of individuals and groups at the most basic level and the stimulation of new kinds of innovations that seek to foster ethical values. As the Ethics Guidelines put it: "Trustworthy AI can improve individual flourishing and collective wellbeing by generating prosperity, value creation and wealth maximization. It can contribute to achieving a fair society, by helping to increase citizens' health and well-being in ways that foster equality in the distribution of economic, social and political opportunity. It is therefore imperative . . . to ensure that everyone can thrive in an AI-based world, and to build a better future while at the same time being globally competitive. . . . We need to make sure these systems are fair in their impact on people's lives, that they are in line with values that should not be compromised and able to act accordingly, and that suitable accountability processes can ensure this. . . . Europe needs to define what normative vision of an AI-immersed future it wants to realise. . . . We intend to contribute to this effort by introducing the notion of Trustworthy AI, which we believe is the right way to build a future with AI. A future where democracy, the rule of law and fundamental rights underpin AI systems and where such systems continuously improve and defend democratic culture will also enable an environment where innovation and responsible competitiveness can thrive. . . . Beyond developing a set of rules, ensuring Trustworthy AI requires us to build and maintain an ethical culture and mind-set through public debate, education and practical learning".[34]

[33] Ibid., at p. 6.

[34] Ibid., at p. 9.

3.6 Fundamental Rights as a Basis for Trustworthy AI

The EU is founded on a constitutional commitment to protect the fundamental and indivisible rights of human beings, to ensure respect for the rule of law, to foster democratic freedom and promote the common good. These rights are reflected in Articles 2 and 3 of the Treaty on European Union, and in the Charter of Fundamental Rights of the EU. The cornerstone values of the EU are the values of respect for human dignity, freedom, democracy, and equality, which are common to the societies of all Member States in which pluralism, non-discrimination, tolerance, justice, solidarity and equality prevail. Furthermore, the EU Charter of Fundamental Rights brings together the personal, civic, political, economic and social rights enjoyed by people within the EU. As it was stated by the High-Level Expert Group (HLEG) on AI, “we believe in an approach to AI ethics based on the fundamental rights enshrined in the EU Treaties, the EU Charter and international human rights law. Respect for fundamental rights, within a framework of democracy and the rule of law, provides the most promising foundations for identifying abstract ethical principles and values, which can be operationalised in the context of AI”.[35] All these fundamental rights are united by reference to a common foundation rooted in respect for human dignity, “hereby reflecting what we describe as a “human-centric approach” in which the human being enjoys a unique and inalienable moral status of primacy in the civil, political, economic and social fields”.[36]

To decide which ethical principles to adopt, the HLEG AI Guidelines looked at specific families of fundamental rights. Among the comprehensive set of indivisible rights set out in the EU Treaties and the EU Charter, five families of fundamental rights—many of which are legally enforceable in the EU so that compliance with their terms is legally obligatory—were found by the HLEG to be particularly apt to cover AI systems. These are the same rights that apply to individuals in liberal democracies making them an ideal blueprint to derive ethical principles for a technology with such far-reaching implications for society. These groups of fundamental rights are the following: (i) respect for human dignity; (ii) freedom of the individual; (iii) respect for democracy, justice and the rule of law; (iv) equality, non-discrimination and solidarity—including the rights of persons at risk of exclusion; (v) citizens’ rights.[37] In particular:

3.6.1 Respect for Human Dignity

Respect for human dignity is the overarching governance instrument for AI-enabled products and services, the value upon which all fundamental rights are grounded.

[35] Ibid.

[36] Ibid., at p. 10.

[37] Ibid., at pp. 10 and 11.

Human dignity is the central value underpinning the entirety of international human rights law and can be understood as the crucial minimum requirement for ensuring the protection of every individuals' self-respect in society. According to the ethics guidelines, "human dignity encompasses the idea that every human being possesses an "intrinsic worth", which should never be diminished, compromised or repressed by others—nor by new technologies like AI systems. In this context, respect for human dignity entails that all people are treated with respect due to them as moral subjects, rather than merely as objects to be sifted, sorted, scored, herded, conditioned or manipulated. AI systems should hence be developed in a manner that respects, serves and protects humans' physical and mental integrity, personal and cultural sense of identity, and satisfaction of their essential needs".[38]

3.6.2 Freedom of the Individual

Human beings should be free to make life decisions for themselves. This also includes intervention from government to ensure that individuals have equal access to AI's benefits and opportunities. Freedom of the individual "means a commitment to enabling individuals to wield even higher control over their lives, including (among other rights) protection of the freedom to conduct a business, the freedom of the arts and science, freedom of expression, the right to private life and privacy, and freedom of assembly and association".[39] For instance, the potential for AI to restrict freedom of movement is directly tied to its use for (unjustified) surveillance. AI can provide a detailed picture of individuals' movements as well as predict future location, especially when embedded in systems that combine data from satellite

[38] Ibid., at p. 10.

C. McCrudden acknowledges the feasibility to distill a minimum content of human dignity that has been applied universally and borrows three elements from Neuman (2000, pp. 249–271) to outline what he calls the 'minimum core' of human dignity. These elements are: (i) every human being possesses an intrinsic worth, merely by being human; (ii) intrinsic worth should be recognized and respected by others, and some forms of treatment are inconsistent with respect for this intrinsic worth; (iii) the state exists for individual human beings (not vice versa). See McCrudden (2008), pp. 655–724.

See also: Hilgendorf (2018), pp. 325 ff; Neuman (2000, pp. 249–271); Zardiashvili and Fosch-Villaronga (2020), pp. 30:121–143; Schroeder and Bani-Sadr (2017); Weisstub (2002), pp. 263–294; Floridi (2016), pp. 307–312; Marchant et al. (2011); Sharkey (2014), pp. 63–75; Floridi (2018), pp. 1–8; Cath (2018), p. 2133; Chopra and White (2011); Dautenhahn et al. (2002); Di Nucci (2017); Dicke (2002); Ebers and Navas Navarro (2020); Fiske et al. (2019); Floridi (2013); Floridi (2014); Floridi (2018); Fosch-Villaronga (2019); Fosch-Villaronga and Albo-Canals (2019); Fosch-Villaronga and Golia (2019); Fosch-Villaronga and Heldeweg (2018); Fosch-Villaronga and Millard (2019); Graeme et al. (2012); Grodzinsky et al. (2008); Guihot et al. (2017); Koops and Leenes (2014); Kritikos (2016); O'Mahony (2012); Tzimas (2021); Veale et al. (2018); Wagner (2018); Winfield and Jirotka (2018).

[39] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at pp. 10–11.

imagery, facial recognition-powered cameras, and cell phone location information. As a result, it could easily be used by governmental authorities to facilitate more precise restriction of the freedom of movement.[40]

3.6.3 Respect for Democracy, Justice and the Rule of Law: Citizens' Rights

AI systems should serve to maintain and encourage democratic processes and respect the plurality of values and life choices of individuals. AI systems must not undermine democratic processes and voting systems. Citizens' rights (including the right to vote and the right to good administration) could be negatively impacted by AI systems and should be safeguarded. For instance, AI-powered surveillance could be used to restrict and inhibit political participation, including by identifying and discouraging certain groups of people from voting. Also, use of facial recognition in polling places or voting booths could compromise the secrecy of the ballot.[41] Also, it is clear that AI systems can be manipulated if personal data is used in a malicious way. We have already seen this unfolding in the US Presidential Elections 2016 where personal data of as many as 50 million Facebook users were acquired—without users' express consent—by Cambridge Analytica in an illicit fashion to create a system that could target crucial US voters in decisive constituencies with tailor-made political ads and other personalized posts based on their psychological profile that made them vote for something they would not have considered otherwise.

3.6.4 Equality, Non-Discrimination and Solidarity: Including the Rights of Persons at Risk of Exclusion

In an AI context, equality means that the system's operations should not generate unfairly biased outputs. For instance, depending on the data input that is used to train AI systems, their outputs could be biased. AI developers should make sure that the design of their algorithms is not biased and that the data used to train AI systems should be as inclusive as possible, representing different population groups. The focus is on avoiding unfair bias when AI products and services are designed and/or

[40] The ability of AI to identify, classify, and discriminate magnifies the potential for human rights abuses in both scale and scope. The role of AI and how it could violate or risk violating human rights—covering also risks posed by prospective future developments in AI—is discussed in "Human Rights in the Age of Artificial Intelligence", Report by Access Now, November 2018, especially at pp. 18–30.

[41] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at p. 11.

used. Just as citizens are equal before the law in democracies, algorithms need to provide the same fairness and be prevented from taking biased decisions that discriminate against certain people as this would cause substantial harm to society.

Equality requires respect for vulnerable persons and groups, such as the elderly, people with disabilities, ethnic minorities, women, children, or others at risk of exclusion.[42] The misuse of AI in this context could occur in various ways. For instance, there could be cases where facial recognition software is not just being used to surveil and identify, but also to target and discriminate. Or when hiring processes have been fraught with bias and discrimination.[43] Other cases include risk scoring (i.e. evaluating whether or not a defendant is likely to reoffend in order to recommend sentencing) or "predictive policing," (i.e. using insights from various data points to predict where or when crime will occur and direct law enforcement action accordingly).[44]

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[42] Ibid.

[43] For instance, a threat could come from automated hiring, where applicants are judged by AI that has learnt from historical data sets. "Discrimination that comes out of systems trained on data from people . . . reflects the behaviour of people who previously carried out the job," explaind Yoshua Bengio, a Professor in the University of Montreal's department of computer science. In response to these concerns, ethical frameworks for AI are being written around the world. The threat of artificial intelligence is not that robots could be like us. The problem, according to scientists, is their inhumanity: we cannot make them care about justice or equality. See "EU backs AI regulation while China and US favour technology", Siddharth Venkataramakrishnan, Financial Times, 25.04.2019, https://www.ft.com/content/4fd088a4-021b-11e9-bf0f-53b8511afd73.

[44] "Human Rights in the Age of Artificial Intelligence", Report by Access Now, November 2018, at p. 24. AI models are designed to sort and filter, whether by ranking search results or categorizing people into buckets. This discrimination can interfere with human rights when it treats different groups of people differently. Sometimes such discrimination has positive social aims, for example, when it is used in programs to promote diversity. In criminal justice, this discrimination is often the result of forms of bias. Use of AI in some systems can perpetuate historical injustice in everything from prison sentencing to loan applications. In 2013, researcher Latanya Sweeney found that a Google search for stereotypically African American-sounding names yielded ads that suggested an arrest record (such as "Trevon Jones, Arrested?") in the vast majority of cases. In 2015, researchers at Carnegie Mellon found Google displayed far fewer ads for high-paying executive jobs to women. Google's personalized ad algorithms are powered by AI, and they are taught to learn from user behavior. The more people click, search, and use the internet in racist or sexist ways, the more the algorithm translates that into ads. This is compounded by discriminatory advertiser preferences and becomes part of a cycle. How people perceive things affects the search results, which affect how people perceive things.

See also: Sweeney (2013); Carpenter (2015).

Given that facial recognition software has higher error rates for darker-skinned faces, it is likely that misidentification will disproportionately affect people of color. The gravity of the problem was demonstrated by the ACLU's test of Amazon's Rekognition facial recognition software. The ACLU scanned the faces of all 535 U.S. members of Congress against 25,000 public criminal mugshots using Rekognition's API with the default 80% confidence level. No one in the U.S. Congress was actually in the mugshot database, yet there were 28 false matches. Of these matches, 38% were people of color, even though only 20% of members of Congress are people of color. This showed that as with many facial recognitions systems, Rekognition disproportionately impacted people of color. See: Brandom (2018).

3.7 Ethical Principles in the Context of AI Systems: The Ethical Principles That Must Be Adhered to in Order to Ensure Ethical and Robust AI

3.7.1 European Group on Ethics in Science and New Technologies ("EGE"): Statement on "Artificial Intelligence, Robotics and 'Autonomous' Systems"

The approach followed by the HLEG's Ethics Guidelines on AI was built on the work undertaken by the European Group on Ethics in Science and New Technologies ("EGE"),[45] and in particular the EGE's Statement on "Artificial Intelligence, Robotics and 'Autonomous' Systems", issued in March 2018,[46] which underlined the risks inherent to uncoordinated approaches in the regulation of AI and autonomous technologies: "Regulatory patchworks may give rise to 'ethics shopping', resulting in the relocation of AI development and use to regions with lower ethical standards. Allowing the debate to be dominated by certain regions, disciplines, demographics or industry actors risks excluding a wider set of societal interests and perspectives".[47]

The EGE recognised that AI, robotics and autonomous systems "can bring prosperity, contribute to well-being and help to achieve European moral ideals and socio-economic goals if designed and deployed wisely"[48] and rejected any concerns that ethics could constitute obstacles to innovation and investment: "ethical considerations and shared moral values are essential to can be used to shape the world of tomorrow and should be construed as stimulus and opportunities for innovation, and not impediments and barriers".[49] As a first step towards the formulation of a set of ethical guidelines that may serve as a basis for the establishment of global standards and legislative action, the EGE's Statement proposed a set of nine (9) fundamental ethical principles, based on the values laid down in the EU Treaties and the EU Charter of Fundamental Rights, that can guide the development of AI. These ethical principles are the following: (a) Human dignity; (b) Autonomy; (c) Responsibility; (d) Justice, equity, and solidarity; (e) Democracy; (f) Rule of law and accountability;

[45] The European Group on Ethics in Science and New Technologies (EGE), founded in 1991, is an independent advisory body of the President of the European Commission, which advises on all aspects of Commission policies where ethical, societal and fundamental rights issues intersect with the development of science and new technologies.

[46] European Group on Ethics in Science and New Technologies (EGE), "Statement on Artificial Intelligence, Robotics and 'Autonomous' Systems", Brussels, 9 March 2018, European Commission, Directorate-General for Research and Innovation ("EGE – Statement on AI, 2018").

[47] Ibid., p. 14.

[48] Ibid., p. 20.

[49] Ibid.

(g) Security, safety, bodily and mental integrity; (h) Data protection and privacy; (i) Sustainability.[50]

Based on these fundamental ethical principles, the EGE's Statement on AI called for a wide-ranging and inclusive public engagement and deliberation on the ethics of AI, robotics and autonomous technology and on the set of values that societies choose to embed in the development and governance of these technologies—"need for a collective, wide-ranging and inclusive process of reflection and dialogue, a dialogue that focuses on the values around which we want to organise society and on the role that technologies should play in it".[51] As a result, this process should provide a platform for bringing together the diverse global initiatives[52] towards the formulation of ethical principles regarding AI and autonomous systems. It should integrate an inclusive societal debate, drawing upon the input of diverse perspectives, where those with different expertise and values can be heard. To this end, the EGE's Statement on AI urged the EU to lead such a process—"calls for the launch of a

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[50] Ibid., pp. 16–19.

[51] Ibid., pp. 5 and 11.

[52] The EGE's Statement on AI mentioned some of the most prominent initiatives towards the formulation of ethical principles regarding AI and autonomous systems, such as the IEEE's (Institute of Electrical and Electronics Engineers) policy paper on 'Ethically Aligned Design' (http://standards.ieee.org/news/2016/ethically_aligned_design.html); ITU's (International Telecommunication Union) Global Summit 'AI for Good' in 2017 (https://www.itu.int/en/ITU-T/AI/Pages/201706-default.aspx); the ACM's (Association for Computing Machinery) work on the issue, including a major AAAI/ACM 'Conference on AI, Ethics, and Society' in February 2018 (http://www.aies-conference.com). The EGE's Statement noted that, within the private sector, companies such as IBM, Microsoft and Google's DeepMind established their own ethic codes on AI and joined forces in creating broad initiatives such as the 'Partnership on AI' (https://www.partnershiponai.org/) or 'OpenAI' (https://openai.com/), which brought together industry, non-profit and academic organisations. One of the leading initiatives calling for a responsible development of AI was launched by the Future of Life Institute and culminated in the creation of the 'Asilomar AI Principles'. This list of 23 fundamental principles to guide AI research and application was signed by hundreds of stakeholders, with signatories representing predominantly scientists, AI researchers and industry (https://futureoflife.org/ai-principles/). A similar participatory process was launched upon the initiative of the Forum on the Socially Responsible Development of Artificial Intelligence held by the University of Montreal in November 2017, in reaction to which a first draft of a potential 'Declaration for a Responsible Development of Artificial Intelligence' was developed. It is publicly accessible on an online platform where all sectors of society are invited to comment on the text (http://nouvelles.umontreal.ca/en/article/2017/11/03/montreal-declaration-for-aresponsible-development-of-artificial-intelligence/). The EGE's Statement also mentioned that the UN has established a special research institute in The Hague to study the governance of Robotics and AI (UNICRI). It also noted, under the aegis of UNESCO, the COMEST Report on robotics ethics (see: UNESCO, 'Report of COMEST on Robotics Ethics', World Commission on the Ethics of Scientific Knowledge and Technology, 2017) and the IBC Report on big data and health, both adopted in September 2017. This was the reason—the lack of a harmonised European approach—which prompted the European Parliament in 2017 to call for a range of measures to prepare for the regulation of advanced robotics, including the development of a guiding ethical framework for the design, production and use of robots (European Parliament, Committee on Legal Affairs 2015/2103 (INL) Report with Recommendations to the Commission on Civil Law Rules on Robotics, Rapporteur Mady Delvaux).

process that paves the way towards a common, internationally recognised ethical and legal framework for the design, production, use and governance of artificial intelligence, robotics, and 'autonomous' systems"–,[53] and called upon the Commission to support its implementation.[54]

The risk of not building an international consensus on AI ethics guidelines was already stressed by the EGE's Statement on AI, which drew attention to the risks inherent to uncoordinated and unbalanced approaches in the regulation of AI.[55] At the same time, this lack of a harmonised approach on AI ethics guidelines—and the subsequent production of various sets of ethical principles—could easily lead to overlap, confusion, ethics shopping and uncertainty. This uncertainty is one of the main reasons why a unified approach is deemed necessary:

> "many organizations have launched a wide range of initiatives to establish ethical principles for the adoption of socially beneficial AI. Unfortunately, the sheer volume of proposed principles threatens to become overwhelming and confusing, posing two potential problems. Either the various sets of ethical principles for AI are similar, leading to unnecessary repetition and redundancy, or, if they differ significantly, confusion and ambiguity will result instead. The worst outcome would be a 'market for principles' where stakeholders may be tempted to 'shop' for the most appealing ones"[56] ... "the increasing demand for reflection and clear policies on the impact of AI on society has yielded a glut of initiatives. Each additional initiative yields a supplementary statement of principles, values, or tenets to guide the development and adoption of AI. The risk is unnecessary repetition and overlap, if the various sets of principles are similar, or confusion and ambiguity, if they differ. In either eventuality, the development of laws, rules, standards, and best practices to ensure that AI is socially beneficial may be delayed by the need to navigate the wealth of principles and declarations set out by an ever-expanding array of initiatives". [57]

3.7.2 The AI4People's Scientific Committee

Against this background, it was important to examine whether the various sets of principles produced by this wide range of initiatives are similar or could be supplementary. To this end, the AI4People's Scientific Committee[58] produced a comparative analysis, including an assessment of whether a unified framework may be synthesised. In particular, the AI4People's Scientific Committee presented a synthesis of existing sets of ethics principles produced by various multi-stakeholder

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[53] European Group on Ethics in Science and New Technologies (EGE), "Statement on Artificial Intelligence, Robotics and 'Autonomous' Systems", Brussels, 9 March 2018, European Commission, Directorate-General for Research and Innovation, p. 20.

[54] Ibid., pp. 15 and 20.

[55] Ibid., p. 14.

[56] "A Unified Framework of Five Principles for AI in Society", by Luciano Floridi and Josh Cowls, Harvard Data Science Review, July 2019 (updated in November 2019), at p. 2.

[57] Ibid., at p. 4.

[58] Floridi et al. (2018), pp. 689–707.

organisations and initiatives,[59] it proposed five ethical principles which should guide the development and deployment of AI in society, and offered twenty recommendations[60]—to assess, to develop, to incentivise, and to support good AI–, with the aim to lay the foundation for the establishment of a "Good AI Society".[61]

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[59]The AI4People's Scientific Committee assessed the following six documents, which, taken together, produced forty-seven (47) ethics principles:

(i) The Asilomar AI Principles, developed under the auspices of the Future of Life Institute, in collaboration with attendees of the high-level Asilomar conference of January 2017 (hereafter "Asilomar"; Asilomar AI Principles 2017), "Principles developed in conjunction with the 2017 Asilomar conference" [Benevolent AI 2017]. See at: https://futureoflife.org/ai-principles/ ["Asilomar AI Principles"].

(ii) The Montreal Declaration for a Responsible Development of Artificial Intelligence, developed under the auspices of the University of Montreal, following the Forum on the Socially Responsible Development of AI of November 2017. Final text, December 2018. The principles mentioned are those which were announced as of 1st May 2018. See at: https://www.montrealdeclaration-responsibleai.com/the-declaration [hereafter "Montreal AI Declaration"]. The Montreal Declaration for responsible AI development has three main objectives: (1) Develop an ethical framework for the development and deployment of AI; (2) Guide the digital transition so everyone benefits from this technological revolution; (3) Open a national and international forum for discussion to collectively achieve equitable, inclusive, and ecologically sustainable AI development.

(iii) The IEEE Initiative on Ethics of Autonomous and Intelligent Systems - Ethically Aligned Design (EAD), 2019. This crowd-sourced global treatise received contributions from 250 global thought leaders to develop principles and recommendations for the ethical development and design of autonomous and intelligent systems. See at: https://ethicsinaction.ieee.org. [hereafter "IEEE Initiative – Ethically Aligned Design"].

(iv) European Group on Ethics in Science and New Technologies, "Statement on Artificial Intelligence, Robotics and 'Autonomous' Systems", March 2018. See at: https://ec.europa.eu/research/ege/pdf/ege_ai_statement_2018.pdf [hereafter "EGE Statement on AI"].

(v) The "five overarching principles for an AI code" offered in paragraph 417 of the UK House of Lords Artificial Intelligence Committee's Report "AI in the UK: ready, willing and able?", published on 16 April 2018 (hereafter "AIUK"; House of Lords 2018). Paragraph 417 of the Report suggested the following "five overarching principles for an AI Code": (1) Artificial intelligence should be developed for the common good and benefit of humanity. (2) Artificial intelligence should operate on principles of intelligibility and fairness. (3) Artificial intelligence should not be used to diminish the data rights or privacy of individuals, families or communities. (4) All citizens have the right to be educated to enable them to flourish mentally, emotionally and economically alongside artificial intelligence. (5) The autonomous power to hurt, destroy or deceive human beings should never be vested in artificial intelligence. See at: https://publications.parliament.uk/pa/ld201719/ldselect/ldai/100/100.pdf [hereafter "AI House of Lords, 2018"].

(vi) "Partnership on AI – Tenets", 2018. Partnership on AI is a multi-stakeholder organisation consisting of academics, researchers, civil society organisations, companies building and utilising AI technology, and other groups. See at: https://partnershiponai.org/about/#tenets [hereafter "Partnership on AI – Tenets, 2018"].

[60]Floridi et al. (2018), pp. 689–707, at pp. 700–705.

[61]As it was explained, if the aim is to create a Good AI Society, the proposed five ethical principles should be embedded in the default practices of AI. To this end, AI should be designed and developed in ways that decrease inequality and further social empowerment, with respect for

The AI4People's Scientific Committee, after examining the sets of ethics principles produced by other organisations and similar initiatives, found "an impressive and reassuring degree of coherence and overlap between the six sets of principles. This can most clearly be shown by comparing the sets of principles with the set of four core principles commonly used in bioethics: beneficence, non-maleficence, autonomy, and justice".[62] The AI4People's Scientific Committee pointed out that, although the aforementioned four bioethical principles "adapt surprisingly well to the fresh ethical challenges posed by artificial intelligence", they are not exhaustive, and stated that a new principle should be added, one which is not drawn from bioethics, namely the principle of explicability, which is conceived as covering both intelligibility and accountability.[63] The authors concluded that an ethical framework for AI, guiding the development and deployment of AI in society, should be formed of the following five ethics principles: (i) Beneficence: Promoting Well-Being, Preserving Dignity, and Sustaining the Planet; (ii) Non-maleficence: Privacy, Security and "Capability Caution"; (iii) Autonomy: The Power to Decide (Whether to Decide); (iv) Justice: Promoting Prosperity and Preserving Solidarity; (v) Explicability: Enabling the Other Principles Through Intelligibility and Accountability.[64] According to the authors, these five principles cover all forty-seven ethics principles which were produced, taken together, in the aforementioned six "high-profile" reports and initiatives.

3.7.3 The Four Ethical Principles Identified by HLEG's Ethics Guidelines for Trustworthy AI

Building on the work undertaken by the European Group on Ethics in Science and New Technologies (EGE) in its "Statement on Artificial Intelligence, Robotics and 'Autonomous' Systems" and responding to its call (as well as relying to a great extent and adopting the main findings and proposals of the AI4People's Scientific Committee), the AI HLEG's Ethics Guidelines identified four ethical principles and their correlated values that must be respected in the development, deployment and

human autonomy, and increase benefits that are shared by all. Also, AI should be explicable, as explicability is considered to be a critical tool for building public trust in the emerging technologies. Furthermore, a multi-stakeholder approach is required, ensuring that AI will serve the needs of society, by enabling developers, users and rule-makers to collaborate from the outset. Different cultural frameworks inform attitudes to new technology. The AI4People's Scientific Committee stated that this report represents a European approach, which is meant to be complementary to other approaches, and underlined its commitment to the development of AI technology in a way that secures people's trust, serves the public interest and strengthens shared social responsibility. Ibid., at p. 701.

[62] Ibid., at p. 696.

[63] Ibid.

[64] Ibid., at pp. 696–700.

use of AI systems. These four ethical principles—specified as “ethical imperatives” that AI practitioners should always try to adhere to—are the following: (i) respect for human autonomy; (ii) prevention of harm; (iii) fairness; (iv) explicability.[65]

(i) The Principle of Respect for Human Autonomy

The introduction of high-tech AI systems and software that exhibit high degrees of what is often referred to as ‘autonomy’, which means that they develop and can function increasingly independently of human operators and without human control (executing tasks that would require intelligence when carried out by humans), requires special reflection and gives rise to a range of crucial—legal and ethical—questions.

Respect for human autonomy is closely associated with the right to human dignity and liberty, as reflected in Articles 1 and 6 of the EU Charter of Fundamental Rights.[66] According to the HLEG’s Ethics Guidelines for Trustworthy AI (HLEG AI Guidelines), humans interacting with AI systems should be able to keep full self-determination over themselves. The principle of respect for human autonomy (which is a classic tenet of bioethics) means that humans should not be deceived, coerced or manipulated by AI systems—the latter should be designed to complement and enhance human cognitive, social and cultural skills.[67] It was stressed that securing human oversight over work processes in AI systems is essential to ensure that AI

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[65] “Ethics Guidelines for Trustworthy AI”, High-Level Expert Group on Artificial Intelligence, 8 April 2019, at pp. 11 and 12.

It should be noted that these ethical principles resemble to the four ethical principles in robotics engineering which were proposed by the European Parliament Resolution of 16 February 2017 with recommendations to the Commission on Civil Law Rules on Robotics (2015/2103(INL)), (2018/C 252/25). In particular, the Draft Code of Ethical Conduct for Robotics Engineers, as an Annex to the European Parliament Resolution, put forward four ethical principles in robotics engineering: (1) beneficence (robots should act in the best interests of humans); (2) non-maleficence (robots should not harm humans); (3) autonomy (human interaction with robots should be voluntary); (4) justice (the benefits of robotics should be distributed fairly). See European Parliament Resolution of 16 February 2017, Annex to the Resolution: Recommendations as to the content of the proposal requested, p. 253. See also point 13 of the EP Resolution: “(13) . . . the guiding ethical framework should be based on the principles of beneficence, non-maleficence, autonomy and justice, on the principles and values enshrined in Article 2 of the Treaty on European Union and in the Charter of Fundamental Rights, such as human dignity, equality, justice and equity, non-discrimination, informed consent, private and family life and data protection, as well as on other underlying principles and values of the Union law, such as non-stigmatisation, transparency, autonomy, individual responsibility and social responsibility, and on existing ethical practices and codes”.

[66] EU Charter of Fundamental Rights: Article 1—Human dignity: Human dignity is inviolable. It must be respected and protected. Article 6—Right to liberty and security: Everyone has the right to liberty and security of person.

[67] “Ethics Guidelines for Trustworthy AI”, High-Level Expert Group on Artificial Intelligence, 8 April 2019, 2019, at p. 12.

systems do not undermine human autonomy or cause other adverse effects. This is the reason why the principle of human autonomy should be central to the AI systems' functionality. As an example, humans should have the right not to be subject to a decision based solely on automated processing when this significantly affects them or produces legal effects.[68]

The key idea is that the allocation of functions between humans and AI systems should follow human-centric design principles and always allow humans to make decisions in their own best interest—"leave meaningful opportunity for human choice".[69] The principle of autonomy and the "allocation of functions between humans and AI systems" were explicitly affirmed in the EGE's Statement on AI where it was argued that "the principle of autonomy implies the freedom of the human being. This translates into human responsibility and thus control over and knowledge about 'autonomous' systems as they must not impair freedom of human beings to set their own standards and norms and be able to live according to them. All 'autonomous' technologies must, hence, honour the human ability to choose whether, when and how to delegate decisions and actions to them. This also involves the transparency and predictability of 'autonomous' systems, without which users would not be able to intervene or terminate them if they would consider this morally required".[70] Moreover, the EGE's Statement on AI highlighted that a crucial issue

[68] See, for instance, Article 22 of Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation—GDPR), which gives individuals the right not to be subject to a decision based solely on automated processing when this produces legal effects on users or similarly significantly affects them: Article 22: Automated individual decision-making, including profiling 1. The data subject shall have the right not to be subject to a decision based solely on automated processing, including profiling, which produces legal effects concerning him or her or similarly significantly affects him or her. 2. Paragraph 1 shall not apply if the decision: (a) is necessary for entering into, or performance of, a contract between the data subject and a data controller; (b) is authorised by Union or Member State law to which the controller is subject and which also lays down suitable measures to safeguard the data subject's rights and freedoms and legitimate interests; or (c) is based on the data subject's explicit consent. 3. In the cases referred to in points (a) and (c) of paragraph 2, the data controller shall implement suitable measures to safeguard the data subject's rights and freedoms and legitimate interests, at least the right to obtain human intervention on the part of the controller, to express his or her point of view and to contest the decision. 4. Decisions referred to in paragraph 2 shall not be based on special categories of personal data referred to in Article 9(1), unless point (a) or (g) of Article 9(2) applies and suitable measures to safeguard the data subject's rights and freedoms and legitimate interests are in place.

[69] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at p. 12.

[70] European Group on Ethics in Science and New Technologies (EGE), "Statement on Artificial Intelligence, Robotics and 'Autonomous' Systems", Brussels, 9 March 2018, European Commission, Directorate-General for Research and Innovation ("EGE - Statement on AI, 2018"), at p. 16.

Furthermore, according to EGE's Statement on AI, "the term 'autonomy' stems from philosophy and refers to the capacity of human persons to legislate for themselves, to formulate, think and choose norms, rules and laws for themselves to follow. It encompasses the right to be free to set one's own standards and choose one's own goals and purposes in life. The cognitive processes that

that stands out in public debate is the use (and misuse) of robotic weapons systems. It pointed out that autonomy should only be attributed to human beings and that, since autonomy is an essential aspect of human dignity, it should not be 'relativized'. Therefore, it concluded, "moral responsibility, in whatever sense, cannot be allocated or shifted to 'autonomous' technology. In recent debates about Lethal Autonomous Weapons Systems (LAWS) . . . there seems to exist a broad consensus that Meaningful Human Control is essential for moral responsibility. The principle of Meaningful Human Control (MHC) was first suggested for constraining the development and utilisation of future weapon systems. This means that humans - and not computers and their algorithms - should ultimately remain in control, and thus be morally responsible".[71]

Similarly, the "Asilomar AI Principles" document endorsed the principle of autonomy, insofar as "humans should choose how and whether to delegate decisions to AI systems, to accomplish human-chosen objectives".[72]

Furthermore, the UK House of Lords Artificial Intelligence Committee's Report adopted the narrower position and explicitly stated that "the autonomous power to

support and facilitate this are among the ones most closely identified with the dignity of human persons and human agency and activity par excellence. They typically entail the features of self-awareness, self-consciousness and self-authorship according to reasons and values. Autonomy in the ethically relevant sense of the word can therefore only be attributed to human beings. . . . autonomy in its original sense is an important aspect of human dignity that ought not to be relativized. Since no smart artefact or system - however advanced and sophisticated - can in and by itself be called 'autonomous' in the original ethical sense, they cannot be accorded the moral standing of the human person and inherit human dignity. Human dignity as the foundation of human rights implies that meaningful human intervention and participation must be possible in matters that concern human beings and their environment. Therefore, in contrast to the automation of production, it is not appropriate to manage and decide about humans in the way we manage and decide about objects or data, even if this is technically conceivable. Such an 'autonomous' management of human beings would be unethical, and it would undermine the deeply entrenched European core values. Human beings ought to be able to determine which values are served by technology, what is morally relevant and which final goals and conceptions of the good are worthy to be pursued. This cannot be left to machines, no matter how powerful they are". See EGE—Statement on AI, 2018, at pp. 9–10.

[71] Ibid., at p. 10. Also: "A second field of contestation and controversy are 'autonomous' weapon systems. These military systems can carry lethal weapons as their payload, but as far as the software is concerned they are not very different from 'autonomous' systems that we could find in a range of civilian domains close to home. A large part of the debate takes place at the Conference on Certain Conventional Weapons in Geneva concerning the moral acceptability of 'autonomous' weapons and legal and moral responsibility for the deployment of these systems. Now attention needs to turn to questions as to what the nature and meaning of 'meaningful human control' over these systems is and how to institute morally desirable forms of control" (at p. 11).

[72] The "Asilomar AI Principles" document, 2017, which comprises a list of 23 fundamental principles to guide AI research and application, was developed under the auspices of the Future of Life Institute, in collaboration with attendees of the high-level Asilomar conference of January 2017 ("Asilomar AI Principles"). See Principle number 16 ("Human Control"), at: https://futureoflife.org/ai-principles/.

hurt, destroy or deceive human beings should never be vested in artificial intelligence".[73]

Also, the Montreal Declaration for a responsible development of AI underlined the need to achieve a balance between human-led and machine-led decision-making, stating that "artificial intelligent systems (AIS) must be developed and used while respecting people's autonomy, and with the goal of increasing people's control over their lives and their surroundings".[74]

In addition, the OECD's Recommendation of the Council on Artificial Intelligence,[75] under the principle titled "human-centred values and fairness", emphasized the importance of respecting the rule of law, human rights and democratic values,

[73] The UK House of Lords Artificial Intelligence Committee's Report "AI in the UK: ready, willing and able?", published on 16 April 2018. See the "five overarching principles for an AI code" offered in paragraph 417 of the Report, especially Principle number (5).

See at: https://publications.parliament.uk/pa/ld201719/ldselect/ldai/100/100.pdf.

[74] "The Montreal Declaration for a Responsible Development of Artificial Intelligence", developed under the auspices of the University of Montreal, following the Forum on the Socially Responsible Development of AI of November 2017. Final text, December 2018 ("Montreal Declaration for AI").

See Principle number 2 ("Respect for Autonomy Principle"), at p. 9, at: https://www.montrealdeclaration-responsibleai.com/the-declaration: "2 – Respect for Autonomy Principle: AIS (artificial intelligent systems) must be developed and used while respecting people's autonomy, and with the goal of increasing people's control over their lives and their surroundings: (1) AIS must allow individuals to fulfill their own moral objectives and their conception of a life worth living. (2) AIS must not be developed or used to impose a particular lifestyle on individuals, whether directly or indirectly, by implementing oppressive surveillance and evaluation or incentive mechanisms. (3) Public institutions must not use AIS to promote or discredit a particular conception of the good life. (4) It is crucial to empower citizens regarding digital technologies by ensuring access to the relevant forms of knowledge, promoting the learning of fundamental skills (digital and media literacy), and fostering the development of critical thinking. (5) AIS must not be developed to spread untrustworthy information, lies, or propaganda, and should be designed with a view to containing their dissemination. (6) The development of AIS must avoid creating dependencies through attention-capturing techniques or the imitation of human characteristics (appearance, voice, etc.) in ways that could cause confusion between AIS and humans".

[75] OECD, Recommendation of the Council on Artificial Intelligence, 2020, OECD/LEGAL/0449, adopted on 22.05.2019. The OECD Recommendation on AI provided the first intergovernmental standard for AI policies and a foundation on which to conduct further analysis and develop tools to support governments in their implementation efforts. The OECD's Recommendation on AI was adopted by the OECD Council at Ministerial level on 22 May 2019 on the proposal of the Committee on Digital Economy Policy (CDEP). The Recommendation aims to foster innovation and trust in AI by promoting the responsible stewardship of trustworthy AI while ensuring respect for human rights and democratic values. Complementing existing OECD standards in areas such as privacy, digital security risk management, and responsible business conduct, the Recommendation focuses on AI-specific issues and sets a standard that is implementable and sufficiently flexible to stand the test of time in this rapidly evolving field. In June 2019, at the Osaka Summit, G20 Leaders welcomed G20 AI Principles, drawn from the OECD Recommendation. The OECD Recommendation on AI identified five complementary values-based principles for the responsible stewardship of trustworthy AI and called on AI actors to promote and implement them. These five ethical principles are the following: (i) inclusive growth, sustainable development and well-being; (ii) human-centred values and fairness; (iii) transparency and explainability; (iv) robustness, security and safety; (v) accountability (at pp. 7–8).

including dignity and autonomy, and stressed that AI stakeholders "should implement mechanisms and safeguards, such as capacity for human determination, that are appropriate to the context and consistent with the state of art".[76]

Moreover, the UNI Global Union Top 10 Principles for Ethical AI,[77] under the principle "adopt a human-in-command approach", stated that "an absolute precondition is that the development of AI must be responsible, safe and useful, where machines maintain the legal status of tools, and legal persons retain control over, and responsibility for, these machines at all times".[78]

The principle of autonomy is not explicitly cited in the IEEE - Ethically Aligned Design (EAD) Report. However, it is recognised that, without autonomy, "humans fail to thrive, create, and innovate" and, therefore, "ethically aligned design should support, not hinder, human autonomy or its expression".[79] Also, it is stressed that "user autonomy is a key and essential consideration that must be taken into account when addressing whether affective systems should be permitted to nudge human beings".[80] Moreover, it is pointed out that operators of AI systems will become less able to question the decisions that algorithms make and, therefore, as the use of AI systems expands, standards for the operators are becoming essential. To this end, and in relation to the principle of autonomy, operators should be in a position to "know when they need to question A/IS and when they need to overrule them. . . . Creators should make provisions for the operators to override A/IS in appropriate circumstances".[81] Furthermore, with the aim to maintain human autonomy over AI

[76] Ibid., at p. 7.

[77] The UNI Global Union Top 10 Principles for Ethical Artificial Intelligence, 2017.

[78] Ibid., at p. 8.

[79] The IEEE (Institute of Electrical and Electronics Engineers) Global Initiative on Ethics of Autonomous and Intelligent Systems (A/IS) - Ethically Aligned Design (EAD) Report: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems (A/IS), 2019 ("IEEE – EAD Report"), at p. 102.

See also the following Recommendations, at p. 102: "1. It is important that human workers' interaction with other workers not always be intermediated by affective systems (or other technology) which may filter out autonomy, innovation, and communication. 2. Human points of contact should remain available to customers and other organizations when using A/IS. 3. Affective systems should be designed to support human autonomy, sense of competence, and meaningful relationships as these are necessary to support a flourishing life. 4. Even where A/IS are less expensive, more predictable, and easier to control than human employees, a core network of human employees should be maintained at every level of decision-making in order to ensure preservation of human autonomy, communication, and innovation. 5. Management and organizational theorists should consider appropriate use of affective and autonomous systems to enhance their business models and the efficacy of their workforce within the limits of the preservation of human autonomy".

[80] Ibid., at p. 100.

[81] Ibid., at p. 32 (see Principle 8 – Competence). According to the IEEE Report, "creators of A/IS should integrate safeguards against the incompetent operation of their systems. Safeguards could include issuing notifications/warnings to operators in certain conditions, limiting functionalities for different levels of operators (e.g., novice vs. advanced), system shut-down in potentially risky conditions, etc.", at pp. 32–33.

systems, the IEEE - Ethically Aligned Design (EAD) Report recommended the following two-step process: "The creation of an ethics-by-design methodology is the first step to addressing human autonomy in A/IS, where a critically applied ethical design of autonomous systems preemptively considers how and where autonomous systems may or may not dissolve human autonomy. The second step is the creation of a pointed and widely applied education curriculum that spans grade school through university, one based on a classical ethics foundation that focuses on providing choice and accountability toward digital being as a priority in information and knowledge societies".[82]

Finally, the AI4People's Scientific Committee reminded that, in a medical context, the principle of autonomy relates to the concept that people should have the right to make decisions for themselves about the treatment they do or not receive; and that this principle of autonomy is impaired when it can be proven that patients do not possess the mental capacity to decide in their own best interests—"autonomy is thus surrendered involuntarily".[83] However, the situation becomes more complex when attempting to apply the principle of autonomy in the field of AI. Indeed, according to the AI4People's Scientific Committee, "when we adopt AI and its smart agency, we willingly cede some of our decision-making power to machines. Thus, affirming the principle of autonomy in the context of AI means striking a balance between the decision-making power we retain for ourselves and that which we delegate to artificial agents".[84]

The AI4People's Scientific Committee, following the analysis provided by the above-mentioned reports and declarations on the principle of autonomy, concluded that it is imperative for humans to always have the option to choose when and how to delegate decisions and actions to AI systems. In order to strike the right balance between the decision-making power—the "allocation of functions between humans and AI systems"—a human-centric design approach should be followed. As it was characteristically stated: "not only should the autonomy of humans be promoted, but also the autonomy of machines should be restricted and made intrinsically reversible, should human autonomy need to be re-established (consider the case of a pilot able

[82] Ibid., at p. 60.

Also, according to the IEEE Report, "ultimately, the behavior of algorithms rests solely in their design, and that design rests solely in the hands of those who designed them. Perhaps more importantly, however, is the matter of choice in terms of how the user chooses to interact with the algorithm. Users often don't know when an algorithm is interacting with them directly or their data which acts as a proxy for their identity. Should there be a precedent for the A/IS user to know when they are interacting with an algorithm? What about consent? The responsibility for the behavior of algorithms remains with the designer, the user, and a set of well-designed guidelines that guarantee the importance of human autonomy in any interaction. As machine functions become more autonomous and begin to operate in a wider range of situations, any notion of those machines working for or against human beings becomes contested. Does the machine work for someone in particular, or for particular groups but not others? Who decides on the parameters? Is it the machine itself? Such questions become key factors in conversations around ethical standards" (at p. 60).

[83] Floridi et al. (2018), p. 698.

[84] Ibid.

to turn of the automatic pilot and regain full control of the airplane). Taken together, the central point is to protect the intrinsic value of human choice—at least for significant decisions—and, as a corollary, to contain the risk of delegating too much to machines. Therefore, what seems most important here is what we might call "meta-autonomy", or a "decide-to-delegate" model: humans should always retain the power to decide which decisions to take, exercising the freedom to choose where necessary, and ceding it in cases where overriding reasons, such as efficacy, may outweigh the loss of control over decision-making. As anticipated, any delegation should remain overridable in principle (deciding to decide again)".[85]

(ii) The Principle of Prevention of Harm (Non-Maleficence)

As already mentioned, AI systems can function increasingly independently of human operators and without human control. At the same time, AI and robots increase their interaction with humans in very diverse fields and the risks posed by these new interactions should be properly tackled, ensuring that a set of safety rules is translated into every stage of contact between AI, robots and humans. Concerns over safety and security related to the use of AI and robots need to be addressed, aiming at ensuring that AI-embedded systems operate in accordance with high safety/security standards. Therefore, AI systems give rise to a range of important questions in relation to safety, security, the prevention of harm and the mitigation of risks. "How can we make a world with interconnected AI and 'autonomous' devices safe and secure and how can we gauge the risks".[86]

According to the prevention of harm principle,[87] "AI systems should neither cause nor exacerbate harm or otherwise adversely affect human beings".[88] Furthermore, the protection of human dignity as well as physical and mental integrity of humans should be ensured. Particular attention should be paid to vulnerable persons and to situations where AI systems can cause adverse impacts due to asymmetries of

[85] Ibid.

[86] European Group on Ethics in Science and New Technologies (EGE), "Statement on Artificial Intelligence, Robotics and 'Autonomous' Systems", Brussels, 9 March 2018, European Commission, Directorate-General for Research and Innovation, at p. 8.

[87] The prevention of harm principle is strongly associated with the protection of physical or mental integrity, as reflected in Article 3 of the EU Charter of Fundamental Rights, which reads as follows: Article 3—Right to integrity of the person: 1. Everyone has the right to respect for his or her physical and mental integrity. 2. In the fields of medicine and biology, the following must be respected in particular: (a) the free and informed consent of the person concerned, according to the procedures laid down by law; (b) the prohibition of eugenic practices, in particular those aiming at the selection of persons; (c) the prohibition on making the human body and its parts as such a source of financial gain; (d) the prohibition of the reproductive cloning of human beings.

[88] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at p. 12.

power or information, such as between employers and employees, businesses and consumers or governments and citizens.[89] It should be also noted that, according to the HLEG's Ethics Guidelines for Trustworthy AI, the prevention of harm principle also covers consideration of the natural environment and all living beings.[90]

Furthermore, AI systems and the environments in which they operate must be robust (both from a technical and social perspective), safe and secure. Society should be confident that AI systems perform in a safe, secure and reliable manner, and that specific safeguards have been foreseen to prevent any unintentional harm. Therefore, technical robustness—which is closely associated with the principle of prevention of harm—is a vital element for achieving Trustworthy AI. "[AI systems] must be technically robust and it should be ensured that they are not open to malicious use".[91] Indeed, AI systems should be technically robust because, even with good intentions, a lack of technological mastery can cause unintentional harm. "Technical robustness requires that AI systems be developed with a preventative approach to risks and in a manner such that they reliably behave as intended while minimising unintentional and unexpected harm, and preventing unacceptable harm. This should also apply to potential changes in their operating environment or the presence of other agents (human and artificial) that may interact with the system in an adversarial manner".[92]

All major reports and initiatives producing recommendations of sets of ethics principles cautioned against the negative consequences of overusing or misusing AI technologies, including the infringement of privacy, and stressed the need to mitigate risks and negative impacts. For instance, with regard to the issue of safety and security of autonomous systems, the EGE's Statement on AI pointed out that, in order to ensure that autonomous systems do not infringe on the human right to bodily and mental integrity and a safe and secure environment, all dimensions of safety must be taken into account by AI developers and AI systems should be strictly tested before their release in the market. The three dimensions of safety are the following: (i) external safety for their environment and users, (ii) reliability and internal robustness, e.g. against hacking, and (iii) emotional safety (mental integrity) with respect to human-machine interaction.[93] It was also pointed out that, with regard to the implications of autonomous systems on private life and privacy, the right to protection of personal information and the right to respect for privacy are crucially challenged. Therefore, autonomous systems should not interfere with the right to

[89] Ibid.

[90] Ibid.

[91] Ibid.

[92] Ibid., at p. 16.

[93] European Group on Ethics in Science and New Technologies (EGE), "Statement on Artificial Intelligence, Robotics and 'Autonomous' Systems", Brussels, 9 March 2018, European Commission, Directorate-General for Research and Innovation, at pp. 18–19. Principle (g) – Security, safety, bodily and mental integrity.

private life, and infringements on personal privacy should be prevented.[94] Yet the infringement of privacy was not considered by the EGE's Statement to be the only danger to be avoided, emphasizing the importance of avoiding the misuse of AI technologies in other ways as well. For instance, it was stated that consideration should be given to issues related to exploitation of vulnerable users by unscrupulous parties as well as to cases linked to asymmetries of power or information.[95] Furthermore, the EGE's Statement on AI stressed the crucial issue of the use (and misuse) of robotic weapons systems (the so-called Lethal Autonomous Weapons Systems - LAWS): "special attention should also be paid to potential dual use and weaponisation of AI, e.g. in cybersecurity, finance, infrastructure and armed conflict".[96]

The Montreal Declaration for a responsible development of AI stated that "every person involved in AI development must exercise caution by anticipating, as far as possible, the adverse consequences of AIS [artificial intelligent systems] use and by taking the appropriate measures to avoid them [the prudence principle]".[97] It also underlined that (i) specific safeguards should be developed in order to prevent or, at least, to limit any unintentional harm; (ii) it is prudent to restrict open access to algorithms when the misuse of an AIS endangers public health or safety; (iii) the risks of user data misuse should be prevented and the integrity and confidentiality of personal data should be always protected; (iv) AI systems, before their release in the market, must meet strict reliability, security, and integrity requirements; (v) it is imperative to ensure that—for the identification and assessment of the risks they might entail—AI systems are strictly tested in real-life scenarios prior to entering the market and that, when subjected to tests, they will not jeopardise people's lives, harm their quality of life, or affect their reputation or psychological integrity.[98]

[94] Ibid., at p. 19: "Principle (h) – Data protection and privacy: In an age of ubiquitous and massive collection of data through digital communication technologies, the right to protection of personal information and the right to respect for privacy are crucially challenged. Both physical AI robots as part of the Internet of Things, as well as AI softbots that operate via the World Wide Web must comply with data protection regulations and not collect and spread data or be run on sets of data for whose use and dissemination no informed consent has been given. 'Autonomous' systems must not interfere with the right to private life which comprises the right to be free from technologies that influence personal development and opinions, the right to establish and develop relationships with other human beings, and the right to be free from surveillance. Also in this regard, exact criteria should be defined and mechanisms established that ensure ethical development and ethically correct application of 'autonomous' systems. In light of concerns with regard to the implications of 'autonomous' systems on private life and privacy, consideration may be given to the ongoing debate about the introduction of two new rights: the right to meaningful human contact and the right to not be profiled, measured, analysed, coached or nudged".

[95] Ibid.

[96] Ibid.

[97] "The Montreal Declaration for a Responsible Development of Artificial Intelligence", 2018 ("Montreal Declaration for AI"). See Principle number 8 ("Prudence Principle"), at p. 15.

[98] Ibid. See Principle number 8 ("Prudence Principle"), at p. 15: "8 – Prudence Principle: Every person involved in AI development must exercise caution by anticipating, as far as possible, the adverse consequences of AIS use and by taking the appropriate measures to avoid them. (1) It is

Furthermore, the UK House of Lords Artificial Intelligence Committee's Report stated that "artificial intelligence should not be used to diminish the data rights or privacy of individuals, families or communities",[99] and highlighted - focusing on personal health data - that "maintaining public trust over the safe and secure use of their data is paramount to the successful widespread deployment of AI".[100]

The Partnership on AI—Tenets emphasized the importance of the prevention of infringement on personal privacy and of avoiding the misuse of AI technologies: "we will work to maximize the benefits and address the potential challenges of AI technologies, by: a) working to protect the privacy and security of individuals. . . . d) ensuring that AI research and technology is robust, reliable, trustworthy, and operates within secure constraints".[101]

The "Asilomar AI Principles" document, apart from the issues of safety ("AI systems should be safe and secure throughout their operational lifetime, and verifiably so where applicable and feasible") and privacy ("people should have the right to access, manage and control the data they generate, given AI systems' power to analyze and utilize that data; . . . the application of AI to personal data must not unreasonably curtail people's real or perceived liberty"), underlined the threats of an

necessary to develop mechanisms that consider the potential for the double use — beneficial and harmful —of AI research and AIS development (whether public or private) in order to limit harmful uses. (2) When the misuse of an AIS endangers public health or safety and has a high probability of occurrence, it is prudent to restrict open access and public dissemination to its algorithm. (3) Before being placed on the market and whether they are offered for charge or for free, AIS must meet strict reliability, security, and integrity requirements and be subjected to tests that do not put people's lives in danger, harm their quality of life, or negatively impact their reputation or psychological integrity. These tests must be open to the relevant public authorities and stakeholders. (4) The development of AIS must preempt the risks of user data misuse and protect the integrity and confidentiality of personal data. (5) The errors and flaws discovered in AIS and SAAD should be publicly shared, on a global scale, by public institutions and businesses in sectors that pose a significant danger to personal integrity and social organization".

[99] The UK House of Lords Artificial Intelligence Committee's Report "AI in the UK: ready, willing and able?", 16 April 2018. See the "five overarching principles for an AI code" offered in paragraph 417 of the Report, especially Principle number (3). The UK House of Lords Artificial Intelligence Committee's Report argued that several issues associated with the use of AI require careful thought, including the issue of determining legal liability, in cases where a decision taken by an algorithm has an adverse impact on someone's life, the potential criminal misuse of AI and data, and the use of AI in autonomous weapons systems. See Chap. 9 of the Committee's Report, Mitigating the risk of Artificial Intelligence, paras 304–347, pp. 95–105.

[100] Ibid., at para. 300, p. 93.

[101] "Partnership on AI – Tenets", 2018. "Tenet number 6: We will work to maximize the benefits and address the potential challenges of AI technologies, by: a) Working to protect the privacy and security of individuals. b) Striving to understand and respect the interests of all parties that may be impacted by AI advances. c) Working to ensure that AI research and engineering communities remain socially responsible, sensitive, and engaged directly with the potential influences of AI technologies on wider society. d) Ensuring that AI research and technology is robust, reliable, trustworthy, and operates within secure constraints. e) Opposing development and use of AI technologies that would violate international conventions or human rights, and promoting safeguards and technologies that do no harm". See at: https://partnershiponai.org/about/#tenets.

AI arms race ("an arms race in lethal autonomous weapons should be avoided"), i.e. the issue of the weaponization of AI, and stated, in the 'longer-term issues' section, that policy-makers "should avoid strong assumptions regarding upper limits on future AI capabilities".[102]

The OECD's Recommendation of the Council on AI, under the principle titled "robustness, security and safety", emphasized the importance of AI systems meeting strict reliability and security requirements and stated that "AI systems should be robust, secure and safe throughout their entire lifecycle so that, in conditions of normal use, foreseeable use or misuse, or other adverse conditions, they function appropriately and do not pose unreasonable safety risk".[103]

The issue of the weaponization of AI and the threats posed by an AI arms race was also highlighted by the UNI Global Union Top 10 Principles for Ethical AI which stated that "lethal autonomous weapons, including cyber warfare, should be banned".[104] It also pointed out that "in the design and maintenance of AI, it is vital that the system is controlled for negative or harmful human-bias".[105]

Moreover, the IEEE—Ethically Aligned Design (EAD) Report pointed out that infringements on personal privacy (listed as part of the "human rights" principle) should be prevented, and stressed the need to avoid deliberate or accidental misuse (such as hacking, misuse of personal data, system manipulation, or exploitation of vulnerable users by unscrupulous parties) and risks.[106] It also argued that creators of

[102] The "Asilomar AI Principles", 2017, developed under the auspices of the Future of Life Institute. See Principles number 6 ("Safety"), number 12 ("Personal Privacy"), number 13 ("Liberty and Privacy"), number 18 ("AI Arms Race"), number 19 ("Capability Caution").

[103] OECD, Recommendation of the Council on Artificial Intelligence, 2020, OECD/LEGAL/0449, adopted on 22.05.2019, at p. 8. See Principle 1.4—Robustness, security and safety: (a) AI systems should be robust, secure and safe throughout their entire lifecycle so that, in conditions of normal use, foreseeable use or misuse, or other adverse conditions, they function appropriately and do not pose unreasonable safety risk. (b) To this end, AI actors should ensure traceability, including in relation to datasets, processes and decisions made during the AI system lifecycle, to enable analysis of the AI system's outcomes and responses to inquiry, appropriate to the context and consistent with the state of art. (c) AI actors should, based on their roles, the context, and their ability to act, apply a systematic risk management approach to each phase of the AI system lifecycle on a continuous basis to address risks related to AI systems, including privacy, digital security, safety and bias.

[104] The UNI Global Union Top 10 Principles for Ethical Artificial Intelligence, 2017, at p. 9.

[105] Ibid., at p. 8.

[106] The IEEE (Institute of Electrical and Electronics Engineers) Global Initiative on Ethics of Autonomous and Intelligent Systems (A/IS)—Ethically Aligned Design (EAD) Report: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems (A/IS), 2019 ("IEEE – EAD Report").

See Principle 1 – Human Rights, at p. 19. "A/IS shall be created and operated to respect, promote, and protect internationally recognized human rights. . . . "human rights", as defined by international law, provide a unilateral basis for creating any A/IS [autonomous and intelligent systems], as these systems affect humans, their emotions, data, or agency".

Principle 3 – Data Agency, at p. 23: "A/IS creators shall empower individuals with the ability to access and securely share their data, to maintain people's capacity to have control over their identity". "In an era where A/IS are already pervasive in society, governments must recognize that limiting the misuse of personal data is not enough".

AI systems should ensure that operators of their technologies have the experience and skills necessary to use these systems safely and appropriately.[107]

The AI4People's Scientific Committee, after summarizing the findings of the above-mentioned reports and initiatives concerning the prevention of harm principle (non-maleficence), concluded that there is an agreement on the need to caution against the negative consequences of overusing or misusing AI technologies, including the infringement of privacy. It stressed that AI systems should be developed in a manner such that they reliably behave as intended, that risks and negative impacts—ranging from technical failures and unintentional mistakes to malicious attacks—should be mitigated and/or prevented (thus covering both overuse/unintentional harm and misuse/deliberate harm) and that the protection of physical and mental integrity of humans should be ensured. As it was pointed out: "From these various warnings, it is not entirely clear whether it is the people developing AI, or the technology itself, which should be encouraged not to do harm—in other words,

Principle 7 – Awareness of Misuse, at p. 31: "A/IS creators shall guard against all potential misuses and risks of A/IS in operation". "New technologies give rise to greater risk of deliberate or accidental misuse, and this is especially true for A/IS. A/IS increases the impact of risks such as hacking, misuse of personal data, system manipulation, or exploitation of vulnerable users by unscrupulous parties. Cases of A/IS hacking have already been widely reported, with driverless cars, for example. The Microsoft Tay AI chatbot was famously manipulated when it mimicked deliberately offensive users. In an age where these powerful tools are easily available, there is a need for a new kind of education for citizens to be sensitized to risks associated with the misuse of A/IS. . . . Responsible innovation requires A/IS creators to anticipate, reflect, and engage with users of A/IS".

[107] Ibid. See Principle 8 – Competence, at p. 32: "Creators shall specify and operators shall adhere to the knowledge and skill required for safe and effective operation". According to the IEEE Report, "while standards for operator competence are necessary to ensure the effective, safe, and ethical application of A/IS, these standards are not the same for all forms of A/IS. The level of competence required for the safe and effective operation of A/IS will range from elementary, such as "intuitive" use guided by design, to advanced, such as fluency in statistics".

The IEEE – EAD Report provided the following recommendations (at pp. 32–33): "1. Creators of A/IS should specify the types and levels of knowledge necessary to understand and operate any given application of A/IS. In specifying the requisite types and levels of expertise, creators should do so for the individual components of A/IS and for the entire systems. 2. Creators of A/IS should integrate safeguards against the incompetent operation of their systems. Safeguards could include issuing notifications/warnings to operators in certain conditions, limiting functionalities for different levels of operators (e.g., novice vs. advanced), system shut-down in potentially risky conditions, etc. 3. Creators of A/IS should provide the parties affected by the output of A/IS with information on the role of the operator, the competencies required, and the implications of operator error. Such documentation should be accessible and understandable to both experts and the general public. 4. Entities that operate A/IS should create documented policies to govern how A/IS should be operated. These policies should include the real-world applications for such A/IS, any preconditions for their effective use, who is qualified to operate them, what training is required for operators, how to measure the performance of the A/IS, and what should be expected from the A/IS. The policies should also include specification of circumstances in which it might be necessary for the operator to override the A/IS. 5. Operators of A/IS should, before operating a system, make sure that they have access to the requisite competencies. The operator need not be an expert in all the pertinent domains but should have access to individuals with the requisite kinds of expertise".

whether it is Frankenstein or his monster against whose maleficence we should be guarding. Confused also is the question of intent: promoting non-maleficence can be seen to incorporate the prevention of both accidental (what we above call "overuse") and deliberate (what we call "misuse") harms arising. In terms of the principle of non-maleficence, this need not be an either/or question: the point is simply to prevent harms arising, whether from the intent of humans or the unpredicted behaviour of machines (including the unintentional nudging of human behaviour in undesirable ways)".[108]

(iii) The Principle of Fairness/Justice

According to the HLEG's Ethics Guidelines for Trustworthy AI, "the development, deployment and use of AI systems must be fair".[109] While it is acknowledged that there could be different interpretations of what fairness means in the context of AI, it appears that the principle of fairness implies a commitment to: (i) using AI in order to correct past injustices such as eliminating unfair discrimination—"ensuring that individuals and groups are free from unfair bias, discrimination and stigmatization"; (ii) preventing the creation of new harms, such as the undermining of existing social structures, and ensuring equal access through inclusive design processes as well as equal treatment (inclusion and diversity throughout the entire AI system's life cycle)—"equal opportunity in terms of access to education, goods, services and technology should be fostered. Moreover, the use of AI systems should never lead to people being deceived or unjustifiably impaired in their freedom of choice"; (iii) distributive justice, i.e. ensuring that the use of AI creates benefits that are shared/shareable—"ensuring equal and just distribution of both benefits and costs".[110]

The HLEG's Ethics Guidelines also stated that AI practitioners, when they are at the stage of selecting means and ends, should respect the principle of proportionality[111] and look at what elements must be included in their decisions so that they can

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[108] Floridi et al. (2018), p. 697.

[109] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at p. 12.

[110] Ibid.

Also, according to the AI4People's Scientific Committee, the principle of justice (which is one of the four classic bioethics principles and includes the aims of promoting prosperity and preserving solidarity) "is typically invoked in relation to the distribution of resources, such as new and experimental treatment options or simply the general availability of conventional healthcare". See Floridi et al. (2018), p. 698.

[111] The principle of proportionality is one of the basic principles of European law and described as follows: "In accordance with the principle of proportionality, which is one of the general principles of Community law, the lawfulness of the prohibition of an economic activity is subject to the condition that the prohibitory measures are appropriate and necessary in order to achieve the objectives legitimately pursued by the legislation in question, it being understood that when there is a choice between several appropriate measures recourse must be had to the least onerous, and the disadvantages caused must not be disproportionate to the aims pursued", see Case C-331/88,

meet their objectives and respect their obligations—"to balance competing interests and objectives".[112] AI practitioners should examine the options available, including an assessment of the anticipated benefits and potential costs of the option selected (i.e. costs which a measure may entail against its prospective benefits), in order to ensure that the selected measure is compatible and consistent with the principle of proportionality. Their decision should incorporate a discussion on the proportionality of the measure selected and include consideration of alternative measures where possible (taking also into account the potential effect of the proposed measure, especially when fundamental rights are affected), so that the least burdensome measure can be implemented—"preference should be given to the one that is least adverse to fundamental rights and ethical norms (e.g. AI developers should always prefer public sector data to personal data)".[113] Therefore, measures taken should be appropriate for attaining the objective pursued, they should not go beyond what is necessary to achieve this objective, and the least onerous measure should be selected: "one should not use a sledge hammer to crack a nut".[114] In addition, the HLEG's Ethics Guidelines for Trustworthy AI pointed out that the principle of fairness involves the ability to contest and seek effective redress against decisions taken by AI systems and by the humans operating them. To this end, it is imperative that the entity accountable for the decision in question should be identifiable, and the decision-making process should be explicable, i.e. a combination of the principles of intelligibility, accountability, transparency, and explicability.[115]

The EGE's Statement on AI, under the principle titled "justice, equity, and solidarity", emphasized the importance of justice and stressed that "AI should contribute to global justice and equal access to the benefits and advantages that

Fedesa and others [1990] ECR I-4023. See also Case C-210/03, judgment of the Court of 14.12.2004, at para. 47: "... the principle of proportionality, which is one of the general principles of Community law, requires that measures implemented through Community provisions are appropriate for attaining the objective pursued and must not go beyond what is necessary to achieve it". See also Case 137/85 *Maizena* [1987] ECR 4587, at para. 15; Case C-339/92 *ADM Olmuhlen* [1993] ECR I-6473, at para. 15; Case C-210/00 *Kaserei Champignon Hofmeister* [2002] ECR I-6453, at para. 59.

[112] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at pp. 12–13.

[113] Ibid. See also at p. 13: "Measures taken to achieve an end (e.g. the data extraction measures implemented to realise the AI optimisation function) should be limited to what is strictly necessary. [The principle of proportionality] also entails that when several measures compete for the satisfaction of an end, preference should be given to the one that is least adverse to fundamental rights and ethical norms (e.g. AI developers should always prefer public sector data to personal data). Reference can also be made to the proportionality between user and deployer, considering the rights of companies (including intellectual property and confidentiality) on the one hand, and the rights of the user on the other".

[114] Ibid., at p. 13.

[115] Ibid. See also at p. 19: "The requirement of accountability complements the above requirements, and is closely linked to the principle of fairness. It necessitates that mechanisms be put in place to ensure responsibility and accountability for AI systems and their outcomes, both before and after their development, deployment and use".

AI, robotics and 'autonomous' systems can bring".[116] It warned against the risk of bias in datasets used to train AI systems, and pointed out that equality means that the system's operations should not generate unfairly biased outputs. Just as all are (or should be) equal before the law, algorithms should provide the same fairness and be prevented from taking biased decisions that discriminate against certain (vulnerable) people or groups of people. To this end, the EGE's Statement on AI noted that AI developers should make sure that the design of their algorithms is not biased, and that the data used to train AI systems should be as inclusive as possible: "discriminatory biases in data sets used to train and run AI systems should be prevented or detected, reported and neutralised at the earliest stage possible".[117]

It also underlined the need to increase societal fairness by ensuring equal opportunity in terms of access to emerging technologies and equal and fair distribution of benefits brought by AI systems. Technological change is transforming our lives and the nature of work at unprecedented speed and, therefore, the EGE's Statement on AI—uniquely among the other reports and initiatives—argued that more action (such as developing new social support models of fair distribution, mutual assistance and benefit sharing) is required in order to defend against threats to "solidarity", to adapt to the changing global environment, to adjust to the challenges posed and reap the benefits of technological advances: "we need a concerted global effort towards equal access to 'autonomous' technologies and fair distribution of benefits and equal opportunities across and within societies. This includes the formulating of new models of fair distribution and benefit sharing apt to respond to the economic transformations caused by automation, digitalisation and AI, ensuring accessibility to core AI technologies, and facilitating training in STEM and digital disciplines, particularly with respect to disadvantaged regions and societal groups. Vigilance is required with respect to the downside of the detailed and massive data on individuals that accumulates and that will put pressure on the idea of solidarity, e.g. systems of mutual assistance such as in social insurance and healthcare. These processes may undermine social cohesion and give rise to radical individualism".[118]

Similarly, the importance of justice is illustrated in the Montreal Declaration for a responsible development of AI, which stated that "the development and use of AIS [artificial intelligence systems] must contribute to the creation of a just and equitable society".[119] In particular, the AI systems' operations should not generate unfairly

[116] European Group on Ethics in Science and New Technologies (EGE), "Statement on Artificial Intelligence, Robotics and 'Autonomous' Systems", Brussels, 9 March 2018, European Commission, Directorate-General for Research and Innovation, at p. 17.

[117] Ibid.

[118] Ibid.

[119] "The Montreal Declaration for a Responsible Development of Artificial Intelligence", 2018. See Principle number 6 ("Equity Principle"), at p. 13: "6 – Equity Principle: The development and use of AIS must contribute to the creation of a just and equitable society. 1. AIS must be designed and trained so as not to create, reinforce, or reproduce discrimination based on — among other things — social, sexual, ethnic, cultural, or religious differences. 2. AIS development must help eliminate relationships of domination between groups and people based on differences of power, wealth, or

biased outputs, i.e. taking biased decisions that discriminate against certain individuals or groups – such as the elderly, people with disabilities, ethnic minorities etc. – as this would cause substantial harm to society ("AIS must be designed and trained so as not to create, reinforce, or reproduce discrimination based on – among other things – social, sexual, ethnic, cultural, or religious differences").[120] Furthermore, it was stated that AI systems' development should (i) tackle abusive—due to asymmetries of power—relationships ("AIS development must help eliminate relationships of domination between groups and people based on differences of power, wealth, or knowledge"),[121] (ii) facilitate equal and fair distribution of benefits brought by AI systems ("AIS development must produce social and economic benefits for all by reducing social inequalities and vulnerabilities"),[122] and (iii) be as inclusive as possible as well as ensure diversity in terms of gender, racial or ethnic origin, disability, age, religion or any other belief ("AI development environments, whether in research or industry, must be inclusive and reflect the diversity of the individuals and groups of the society").[123]

The UK House of Lords Artificial Intelligence Committee's Report recognised that "the risk of greater societal and regional inequalities emerging as a consequence of the adoption of AI and advances in automation is very real",[124] it stressed that modernisation of education should be a priority and pointed out that specific measures in education and training should be adopted so that all citizens can have every opportunity to acquire the skills they need and thus benefit from AI and digital

knowledge. 3. AIS development must produce social and economic benefits for all by reducing social inequalities and vulnerabilities. 4. Industrial AIS development must be compatible with acceptable working conditions at every step of their life cycle, from natural resources extraction to recycling, and including data processing. 5. The digital activity of users of AIS and digital services should be recognized as labor that contributes to the functioning of algorithms and creates value. 6. Access to fundamental resources, knowledge and digital tools must be guaranteed for all. 7. We should support the development of commons algorithms — and of open data needed to train them — and expand their use, as a socially equitable objective".

[120] Ibid., Principle number 6 ("Equity Principle"), at p. 13, at 6.1.

[121] Ibid., Principle number 6 ("Equity Principle"), at p. 13, at 6.2.

[122] Ibid., Principle number 6 ("Equity Principle"), at p. 13, at 6.3.

[123] Ibid., Principle number 7 ("Diversity - Inclusion Principle"), at p. 14, at 7.3.

[124] The UK House of Lords Artificial Intelligence Committee's Report "AI in the UK: ready, willing and able?", 16 April 2018, at para. 275, p. 86. Also: "The UK must be ready for the disruption that AI will have on the way in which we work. We support the Government's interest in developing adult retraining schemes, as we believe that AI will disrupt a wide range of jobs over the coming decades, and both blue- and white-collar jobs which exist today will be put at risk. It will therefore be important to encourage and support workers as they move into the new jobs and professions we believe will be created as a result of new technologies, including AI. The National Retraining Scheme could play an important role here, and must ensure that the recipients of retraining schemes are representative of the wider population. Industry should assist in the financing of the National Retraining Scheme by matching Government funding. This partnership would help improve the number of people who can access the scheme and better identify the skills required" (at para. 236, p. 76 of the Report).

innovation.[125] It underlined that everyone must have access to the opportunities provided by AI[126] and that all citizens should be entitled "to be educated to enable them to flourish mentally, emotionally and economically alongside artificial intelligence".[127] It also argued that AI systems should be designed and trained in a way that no unfairly biased outputs are generated,[128] i.e. taking biased decisions that discriminate against certain individuals or groups, that they should be inclusive and reflect the diversity of the individuals and groups of the society (the same should apply for the researchers and developers teams, which should be diverse and representative of wider society), and thus reduce social inequalities.[129] As it was stated by the UK House of Lords Artificial Intelligence Committee's Report, in relation to the risk of prejudice in those decisions: "We are concerned that many of the datasets currently being used to train AI systems are poorly representative of the wider population, and AI systems which learn from this data may well make unfair decisions which reflect the wider prejudices of societies past and present. While many researchers, organisations and companies developing AI are aware of these

[125] Ibid. See, for instance, at para. 238, p. 77 of the Report: "Education and artificial intelligence – Artificial intelligence, regardless of the pace of its development, will have an impact on future generations. The education system needs to ensure that it reflects the needs of the future, and prepares children for life with AI and for a labour market whose needs may well be unpredictable. Education in this context is important for two reasons. First, to improve technological understanding, enabling people to navigate an increasingly digital world, and inform the debate around how AI should, and should not, be used. Second, to ensure that the UK can capitalise on its position as a world leader in the development of AI, and grow this potential".

[126] Ibid., at para. 276, p. 86 of the Report. "Everyone must have access to the opportunities provided by AI. The Government must outline its plans to tackle any potential societal or regional inequality caused by AI, and this must be explicitly addressed as part of the implementation of the Industrial Strategy".

[127] Ibid., see the "five overarching principles for an AI code" offered in paragraph 417 of the Report, especially Principle number (4).

[128] Amongst the immediate solutions offered by witnesses to tackle the issue of bias was the creation of more diverse datasets, which could fairly reflect the societies and communities which AI systems are increasingly affecting. Other witnesses highlighted the need to remove bias from training data and the AI systems developed using them. However, it was also argued that this is more complicated. As Dr Ing. Konstantinos Karachalios, Managing Director of IEEE-Standards Association stated, "you can never be neutral; it is us. This is projected in what we do. It is projected in our engineering systems and algorithms and the data that we are producing. The question is how these preferences can become explicit, because if it can become explicit it is accountable and you can deal with it. If it is presented as a fact, it is dangerous; it is a bias and it is hidden under the table and you do not see it. It is the difficulty of making implicit things explicit". Ibid., at para. 110, p. 41 of the Report.

[129] On the other hand, the UK House of Lords Artificial Intelligence Committee Report pointed out that, according to several witnesses, AI could help tackle long-standing prejudices and social inequalities. For instance, as Kriti Sharma, Vice-president of Artificial Intelligence and Bots, Sage, argued: "AI can help us fix some of the bias as well. Humans are biased; machines are not, unless we train them to be. AI can do a good job at detecting unconscious bias as well. For example, if feedback is given in performance reviews where different categories of people are treated differently, the machine will say, 'That looks weird. Would you like to reconsider that?". Ibid., at para. 118, p. 44.

issues, and are starting to take measures to address them, more needs to be done to ensure that data is truly representative of diverse populations, and does not further perpetuate societal inequalities. Researchers and developers need a more developed understanding of these issues. In particular, they need to ensure that data is preprocessed to ensure it is balanced and representative wherever possible, that their teams are diverse and representative of wider society, and that the production of data engages all parts of society. Alongside questions of data bias, researchers and developers need to consider biases embedded in the algorithms themselves—human developers set the parameters for machine learning algorithms, and the choices they make will intrinsically reflect the developers' beliefs, assumptions and prejudices. The main ways to address these kinds of biases are to ensure that developers are drawn from diverse gender, ethnic and socio-economic backgrounds, and are aware of, and adhere to, ethical codes of conduct".[130] "We recommend that a specific

[130] Ibid., at paras 119–120, p. 44. See also paras 107–118 ("Addressing prejudice") of the UK House of Lords Artificial Intelligence Committee's Report: "The current generation of AI systems, which have machine learning at their core, need to be taught how to spot patterns in data, and this is normally done by feeding them large bodies of data, commonly known as training datasets. These systems are designed to spot patterns, and if the data is unrepresentative, or the patterns reflect historical patterns of prejudice, then the decisions which they make may be unrepresentative or discriminatory as well. This can present problems when these systems are relied upon to make real world decisions. Within the AI community, this is commonly known as 'bias'. While the term 'bias' might at first glance appear straightforward, there are in fact a variety of subtle ways in which bias can creep into a system. Much of the data we deem to be useful is about human beings, and is collected by human beings, with all of the subjectivity that entails. As LexisNexis UK put it, "biases may originate in the data used to train the system, in data that the system processes during its period of operation, or in the person or organisation that created it. There are additional risks that the system may produce unexpected results when based on inaccurate or incomplete data, or due to any errors in the algorithm itself". It is also important to be aware that bias can emerge when datasets inaccurately reflect society, but it can also emerge when datasets accurately reflect unfair aspects of society. For example, an AI system trained to screen job applications will typically use datasets of previously successful and unsuccessful candidates, and will then attempt to determine particular shared characteristics between the two groups to determine who should be selected in future job searches. While the intention may be to ascertain those who will be capable of doing the job well, and fit within the company culture, past interviewers may have consciously or unconsciously weeded out candidates based on protected characteristics (such as age, sexual orientation, gender, or ethnicity) and socio-economic background, in a way which would be deeply unacceptable today". [paras 107–109 of the Report].

Several witnesses pointed out that the issue could not be easily confined to the datasets themselves, and in some cases "bias and discrimination may emerge only when an algorithm processes particular data". The Centre for the Future of Intelligence emphasised that "identifying and correcting such biases poses significant technical challenges that involve not only the data itself, but also what the algorithms are doing with it (for example, they might exacerbate certain biases, or hide them, or even create them)". The difficulties in fixing these issues was illustrated recently when it emerged that Google has still not fixed its visual identification algorithms, which could not distinguish between gorillas and black people, nearly three years after the problem was first identified. Instead, Google has simply disabled the ability to search for gorillas in products such as Google Photos which use this feature. The consequences of this are already starting to be felt. Several witnesses highlighted the growing use of AI within the US justice system, in particular the Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) system, challenge be established within the Industrial Strategy Challenge Fund to stimulate the creation of authoritative tools and systems for auditing and testing training datasets to ensure they are representative of diverse populations, and to ensure that when used to train AI systems they are unlikely to lead to prejudicial decisions. This challenge should be established immediately, and encourage applications by spring 2019. Industry must then be encouraged to deploy the tools which are developed and could, in time, be regulated to do so".[131]

The Asilomar AI Principles document underlined the need for "shared benefit" and "shared prosperity" stemming from AI. In particular, it stated that AI development should produce benefits for all ("Shared Benefit - AI technologies should benefit and empower as many people as possible")[132] and that, in order to ensure inclusiveness and reduce social and economic inequalities, these benefits should be equally and fairly distributed: "Shared Prosperity - The economic prosperity created by AI should be shared broadly, to benefit all of humanity".[133]

Likewise, the Partnership on AI—Tenets emphasized the importance of societal fairness by ensuring equal opportunity and fair distribution of benefits brought by AI systems: "we will seek to ensure that AI technologies benefit and empower as many people as possible";[134] "we will work to maximize the benefits and address the potential challenges of AI technologies, by striving to understand and respect the interests of all parties that may be impacted by AI advances".[135]

In the OECD's Recommendation of the Council on AI, the principle of justice is cited under the principles titled 'inclusive growth, sustainable development and well-being', and 'human-centred values and fairness'. In particular, the OECD's Recommendation on AI underlined that AI technologies should ensure inclusiveness and reduce social and economic inequalities ("advancing inclusion of underrepresented populations, reducing economic, social, gender and other inequalities"), and emphasized that AI stakeholders should respect the law and human rights, including

developed by Northpointe, and used across several US states to assign risk ratings to defendants, which help to assist judges in sentences and parole decisions. Will Crosthwait, co-founder of AI start-up Kensai, highlighted investigations which found that the system commonly overestimated the recidivism risk of black defendants, and underestimated that of white defendants. Big Brother Watch observed that here in the UK, Durham Police have started to investigate the use of similar AI systems for determining whether suspects should be kept in custody, and described this and other developments as a "very worrying trend, particularly when the technology is being trialled when its abilities are far from accurate". Evidence from Sheena Urwin, Head of Criminal Justice at Durham Constabulary, emphasised the considerable lengths that Durham Constabulary have taken to ensure their use of these tools is open, fair and ethical, in particular the development of their 'ALGO-CARE' framework for the ethical use of algorithms in policing. [paras 112–113 of the Report].

[131] Ibid., at para. 121, p. 44.

[132] The "Asilomar AI Principles" document, 2017, developed under the auspices of the Future of Life Institute. See Principle number 14 ("Shared Benefit").

[133] Ibid., Principle number 15 ("Shared Prosperity").

[134] "Partnership on AI – Tenets", 2018. Tenet number 1.

[135] Ibid., Tenet number 6(b).

"non-discrimination and equality, diversity, fairness, social justice, and internationally recognised labour rights".[136]

Similarly, the UNI Global Union Top 10 Principles for Ethical AI argued that AI systems should be designed in a way that no unfairly biased outputs are generated ("in the design and maintenance of AI, it is vital that the system is controlled for negative or harmful human-bias, and that any bias – be it gender, race, sexual orientation, age, etc. – is identified and is not propagated by the system").[137] It also stressed that, in order to ensure inclusiveness, all should have access to the opportunities and benefits provided by AI development and that those benefits should be equally and fairly shared: "AI technologies should benefit and empower as many people as possible. The economic prosperity created by AI should be distributed broadly and equally, to benefit all of humanity. Global as well as national policies aimed at bridging the economic, technological and social digital divide are therefore necessary".[138]

The principle of justice or fairness is not explicitly cited in the IEEE - Ethically Aligned Design (EAD) Report. However, in Section 1, under the title "A/IS in Service to Sustainable Development for All", it is underlined that equality should apply both within and among nations. AI technology should serve humanity by improving the quality of life, spreading the benefits brought by the emerging technologies equally for all people, and promoting social justice. It is stressed that equal access to AI-related technology should be ensured in order to drive the well-being of all people, i.e. ensuring autonomous and intelligent systems "for the common good". It is argued that equality requires respect for vulnerable persons and groups or others at risk of exclusion, that equal opportunities in terms of access to digital emerging technologies and fair distribution ("contributing to shared prosperity") of benefits brought by AI systems should be ensured, and that justice, equality, non-discrimination, diversity, and inclusiveness should be promoted. As the IEEE - Ethically Aligned Design (EAD) Report stated: "A/IS will potentially disrupt economic, social, and political relationships and interactions at many levels. Those disruptions could provide an historical opportunity to reset those relationships in order to distribute power and wealth more equitably and thus promote social justice. They could also leverage quality and better standards of life and protect people's dignity, while maintaining cultural diversity and protecting the environment";[139] "the ethical imperative driving this chapter is that A/IS must be harnessed to benefit humanity, promote equality, and realize the world community's vision of a

[136] OECD, Recommendation of the Council on Artificial Intelligence, 2020, OECD/LEGAL/0449, adopted on 22.05.2019, at p. 7.

[137] The UNI Global Union Top 10 Principles for Ethical Artificial Intelligence, 2017. Principle 5 - Ensure a Genderless, Unbiased AI, at p. 8.

[138] Ibid., Principle 6—Share the Benefits of AI Systems, at p. 8.

[139] The IEEE (Institute of Electrical and Electronics Engineers) Global Initiative on Ethics of Autonomous and Intelligent Systems (A/IS) - Ethically Aligned Design (EAD) Report: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems (A/IS), 2019 ("IEEE – EAD Report"), at p. 140.

sustainable future";[140] "we recognize that how A/IS are deployed globally will be a determining factor in whether, in fact, "no-one gets left behind", whether human rights and dignity of all people are respected, whether children are protected, and whether the gap between rich and poor, within and between nations, narrows or widens";[141] "for example, A/IS create the risk of accelerating inequality within and among nations, if their development and marketing are controlled by a few select companies, primarily in HIC. The benefits would largely accrue to the highly educated and wealthier segment of the population, while displacing the less educated workforce, both by automation and by the absence of educational or retraining systems capable of imparting skills and knowledge needed to work productively alongside A/IS. . . . The inequality in accessing and using the internet, both within and among countries, raises questions on how to spread A/IS benefits across humanity. Ensuring A/IS "for the common good" is an ethical imperative and at the core of Ethically Aligned Design, First Edition; the key elements of this "common good" are that it is human-centered, accountable, and ensure outcomes that are fair and inclusive";[142] "this chapter explores the imperative for A/IS to serve humanity by improving the quality and standard of life for all people everywhere. It makes recommendations for advancing equal access to this transformative technology, so that it drives the well-being of all people, rather than further concentrating wealth, resources, and decision-making power in the hands of a few countries, companies, or citizens".[143]

(iv) The Principle of Explicability

The HLEG's Ethics Guidelines for Trustworthy AI, influenced by the work of the AI4People's Scientific Committee,[144] stated that "explicability is crucial for building and maintaining users' trust in AI systems".[145] It pointed out that it is not always possible to provide an explanation as to why an AI system has produced a particular output or decision ('black box' algorithms cases, where explicability measures such as traceability, auditability and transparent communication on system capabilities may be required)[146]—"AI's workings are often invisible or unintelligible to all but (at best) the most expert observers"[147]—and, to this end, an increase in transparency

[140] Ibid.
[141] Ibid., at p. 141.
[142] Ibid.
[143] Ibid.
[144] Floridi et al. (2018), especially at pp. 699–700.
[145] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at p. 13.
[146] Ibid.
[147] Floridi et al. (2018), p. 700.

and the subsequent traceability of AI systems should be ensured.[148] Therefore, the data sets and the processes that generate the AI system's decision, including those of data gathering, the algorithms used and the decision itself, should be well documented. Traceability, apart from fostering transparency, also facilitates explainability, i.e. assists to detect the reasons why a decision taken by an AI system was erroneous which, in turn, could help avoid making similar mistakes in the future.[149] Moreover, the algorithmic decision-making process followed by the AI system (including design choices of the system, as well as the rationale for deploying it) should be—to the extent possible—suitably explainable to the persons affected. To put it simply, explainability is linked with the ability to explain both the technical processes of an AI system and the related human decisions and, therefore, the decisions made by an AI system should be understood, traced and duly contested by humans;[150] "the need to understand and hold to account the decision-making processes of AI".[151]

In addition, it was stressed that the AI system's capabilities and limitations should be adequately and appropriately communicated to the different stakeholders involved, that users should have always the right to be informed that they are interacting with an AI system (and which persons are responsible for it) and, therefore, AI systems should be identifiable as such and not misinform and mislead users by presenting themselves as humans.[152] Finally, the HLEG's Ethics Guidelines for Trustworthy AI pointed out that the principle of proportionality should apply and argued that the extent to which explicability is required should depend on the context and the severity of the consequences (e.g. high-risk uses of the technology, which are likely to arise where AI applications could affect fundamental rights or the legal rights of individuals or pose risk of death, injury or significant material or immaterial damage) if the output or decision generated by an AI system is erroneous or inaccurate. For instance, there should be a much lower tolerance for errors in healthcare that could place patients at risk of adverse outcomes and, therefore, transparency and validation for AI applications designed for healthcare should be imperative/obligatory. As the HLEG's Ethics Guidelines put it, "little ethical concern may flow from inaccurate shopping recommendations generated by an AI

[148] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at p. 13.

[149] Ibid., at pp. 13 and 18. See also Commission Communication, "Building Trust in Human-Centric Artificial Intelligence", COM(2019) 168 final, Brussels, 8.4.2019, at pp. 5–6.

[150] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at pp. 13 and 18. See also Commission Communication, "Building Trust in Human-Centric Artificial Intelligence", COM(2019) 168 final, Brussels, 8.4.2019, at pp. 5–6.

[151] Floridi et al. (2018), p. 700.

[152] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at pp. 13 and 18. See also Commission Communication, "Building Trust in Human-Centric Artificial Intelligence", COM(2019) 168 final, Brussels, 8.4.2019, at pp. 5–6.

system, in contrast to AI systems that evaluate whether an individual convicted of a criminal offence should be released on parole".[153]

In the EGE's Statement on AI the explicability principle was expressed under the terms "transparency" and "accountability", which are used to describe the necessity to explain and to hold to account the decision-making processes of an AI system. It was recognised that it is not always feasible to explain the reasons why an AI system has produced a particular output or decision, especially because "deep learning and so-called 'generative adversarial network approaches' enable machines to 'teach' themselves new strategies and look for new evidence to analyse. In this sense, their actions are often no longer intelligible, and no longer open to scrutiny by humans. This is the case because, first, it is impossible to establish how they accomplish their results beyond the initial algorithms. Second, their performance is based on the data that have been used during the learning process and that may no longer be available or accessible. Thus, biases and errors that they have been presented with in the past become engrained into the system".[154] Thus "AI's inner workings can be extremely hard – if not impossible – to track, explain and critically evaluate".[155] The EGE's Statement on AI pointed out that there are serious questions about the explainability and transparency of AI and autonomous systems,[156] it asked whether—and how—moral responsibility should be attributed and apportioned and who should be held responsible (and in what sense) for erroneous or inaccurate outcomes,[157] and underlined that "the principle of responsibility must be fundamental to AI research and application".[158]

Similarly, the Asilomar AI Principles document described the explicability principle under the terms "transparency" and "responsibility": "if an AI system causes harm, it should be possible to ascertain why";[159] "any involvement by an autonomous system in judicial decision-making should provide a satisfactory explanation auditable by a competent human authority";[160] "designers and builders of advanced AI systems are stakeholders in the moral implications of their use, misuse, and actions, with a responsibility and opportunity to shape those implications".[161]

In the Montreal Declaration for a responsible development of AI, the explicability principle is expressed under the principles 'democratic participation' ("AIS must

[153] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at p. 13.

[154] European Group on Ethics in Science and New Technologies (EGE), "Statement on Artificial Intelligence, Robotics and 'Autonomous' Systems", Brussels, 9 March 2018, European Commission, Directorate-General for Research and Innovation ("EGE - Statement on AI, 2018"), at p. 6.

[155] Ibid., at p. 7.

[156] Ibid., at p. 8.

[157] Ibid.

[158] Ibid., at pp. 16–17.

[159] The "Asilomar AI Principles" document, 2017, developed under the auspices of the Future of Life Institute. See Principle number 7 ("Failure Transparency").

[160] Ibid., Principle number 8 ("Judicial Transparency").

[161] Ibid., Principle number 9 ("Responsibility").

meet intelligibility, justifiability, and accessibility criteria, and must be subjected to democratic scrutiny, debate, and control")[162] and 'responsibility' ("the development and use of AIS must not contribute to lessening the responsibility of human beings when decisions must be made"),[163] which describe the necessity to explain and to hold to account the decision-making processes of an AI system.

Moreover, the Partnership on AI—Tenets used the terms 'understandable' and 'interpretable': "we believe that it is important for the operation of AI systems to be understandable and interpretable by people, for purposes of explaining the technology".[164]

In the OECD's Recommendation of the Council on AI, the explicability principle is cited under the principles titled 'transparency and explainability' and 'accountability'. In particular, the OECD's Recommendation on AI stressed that the AI

[162] "The Montreal Declaration for a Responsible Development of Artificial Intelligence", December 2018.

Principle number 5 ("Democratic Participation Principle"), at p. 12: "5 – Democratic Participation Principle: AIS must meet intelligibility, justifiability, and accessibility criteria, and must be subjected to democratic scrutiny, debate, and control. 1. AIS processes that make decisions affecting a person's life, quality of life, or reputation must be intelligible to their creators. 2. The decisions made by AIS affecting a person's life, quality of life, or reputation should always be justifiable in a language that is understood by the people who use them or who are subjected to the consequences of their use. Justification consists in making transparent the most important factors and parameters shaping the decision, and should take the same form as the justification we would demand of a human making the same kind of decision. 3. The code for algorithms, whether public or private, must always be accessible to the relevant public authorities and stakeholders for verification and control purposes. 4. The discovery of AIS operating errors, unexpected or undesirable effects, security breaches, and data leaks must imperatively be reported to the relevant public authorities, stakeholders, and those affected by the situation. 5. In accordance with the transparency requirement for public decisions, the code for decision-making algorithms used by public authorities must be accessible to all, with the exception of algorithms that present a high risk of serious danger if misused. 6. For public AIS that has a significant impact on the life of citizens, citizens should have the opportunity and skills to deliberate on the social parameters of these AIS, their objectives, and the limits of their use. 7. We must at all times be able to verify that AIS are doing what they were programed for and what they are used for. 8. Any person using a service should know if a decision concerning them or affecting them was made by an AIS. 9. Any user of a service employing chatbots should be able to easily identify whether they are interacting with an AIS or a real person. 10. Artificial intelligence research should remain open and accessible to all".

[163] Ibid. Principle number 9 ("Responsibility Principle"), especially paras 1 and 5, at p. 16: "9 – Responsibility Principle: The development and use of AIS must not contribute to lessening the responsibility of human beings when decisions must be made. 1. Only human beings can be held responsible for decisions stemming from recommendations made by AIS, and the actions that proceed therefrom. 2. In all areas where a decision that affects a person's life, quality of life, or reputation must be made, where time and circumstance permit, the final decision must be taken by a human being and that decision should be free and informed. 3. The decision to kill must always be made by human beings, and responsibility for this decision must not be transferred to an AIS. 4. People who authorize AIS to commit a crime or an offense, or demonstrate negligence by allowing AIS to commit them, are responsible for this crime or offense. 5. When damage or harm has been inflicted by an AIS, and the AIS is proven to be reliable and to have been used as intended, it is not reasonable to place blame on the people involved in its development or use".

[164] "Partnership on AI – Tenets", 2018, see Tenet number 7.

system's capabilities should be communicated to the different stakeholders involved, that users should have the right to be informed that they are interacting with an AI system, and that the decisions made by an AI system should be understood, traced and, if needed, contested by humans: "AI Actors should commit to transparency and responsible disclosure regarding AI systems. To this end, they should provide meaningful information, appropriate to the context, and consistent with the state of art: (i) to foster a general understanding of AI systems, (ii) to make stakeholders aware of their interactions with AI systems, including in the workplace, (iii) to enable those affected by an AI system to understand the outcome, and, (iv) to enable those adversely affected by an AI system to challenge its outcome based on plain and easy-to-understand information on the factors, and the logic that served as the basis for the prediction, recommendation or decision".[165] Moreover, the OECD's Recommendation on AI underlined that accountability should be ensured in order to be able to identify who should be responsible—and held liable—in the event of failure of an AI system and, subsequent, harm ("AI actors should be accountable for the proper functioning of AI systems and for the respect of the above principles, based on their roles, the context, and consistent with the state of art").[166]

The IEEE - Ethically Aligned Design (EAD) Report made specific reference to the principles of 'transparency' ("the basis of a particular A/IS decision should always be discoverable") and 'accountability' ("A/IS shall be created and operated to provide an unambiguous rationale for all decisions made").[167]

In particular, the IEEE Report highlighted as a "key concern" the lack of transparency in the operation of AI and autonomous systems.[168] According to the Report, transparent are the AI systems in which it is possible to discover how and why a particular decision was taken or, in the case of a robot, why it acted the way it did. The term 'transparency' is used in a way that also covers the concepts of traceability, explainability, and interpretability.[169] The IEEE Report acknowledged that lack of transparency in specific AI systems—especially the safety critical systems, i.e. those that can raise the potential level of harm (e.g. medical diagnosis or driverless car autopilots)—increases the risk and extent of damage since it may make it extremely difficult for humans to understand the actions of the AI systems that they use or, in case of a failure, to fix faults following accidents. Therefore, a lack of transparency and the inability to scrutinise the workings of deep learning systems could lead to lack of accountability which denies those who are harmed by AI-informed decisions adequate access to justice or effective remedies. As it was

[165] OECD, Recommendation of the Council on Artificial Intelligence, 2020, OECD/LEGAL/0449, adopted on 22.05.2019, at p. 8.

[166] Ibid.

[167] The IEEE (Institute of Electrical and Electronics Engineers) Global Initiative on Ethics of Autonomous and Intelligent Systems (A/IS) - Ethically Aligned Design (EAD) Report: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems (A/IS), 2019, Principles 5 (Transparency) and 6 (Accountability).

[168] Ibid., Principle 5 (Transparency), at p. 27.

[169] Ibid.

put, lack of transparency "will complicate efforts to determine and allocate responsibility when something goes wrong"; "it will also increase the difficulty of ensuring accountability".[170] The IEEE Report argued that the operation of AI systems should be transparent to a wide range of stakeholders for different reasons[171] and concluded that new standards (which would describe measurable, testable levels of transparency) should be developed, so that AI systems can be objectively evaluated and levels of compliance determined: "The mechanisms by which transparency is provided will vary significantly, including but not limited to, the following use cases: 1. For users of care or domestic robots, a "why-did-you-do-that button" which, when pressed, causes the robot to explain the action it just took. 2. For validation or certification agencies, the algorithms underlying the A/IS and how they have been verified. 3. For accident investigators, secure storage of sensor and internal state data comparable to a flight data recorder or black box".[172]

As regards the principle of accountability, the IEEE Report stated that the programming, output, and purpose of AI systems are often not visible and detectable by the general public and, therefore, AI systems should be created and operated to provide a clear explanation for decisions made.[173] It pointed out that "accountability is not possible without transparency", thus this principle is closely associated with the principle of transparency.[174] It argued that, in order to increase transparency and minimise the risk of bias or error, AI systems should be developed in a manner which allows humans to understand the basis of their actions.[175] Also, clarity concerning the manufacture and deployment of AI systems should be ensured in order to establish responsibility and accountability, and to prevent potential harm. Indeed, in a system with multiple stakeholders and decisions being made by AI, it will be important to identify who should be responsible—and held liable—in the event of failure: "manufacturers of these systems must be accountable in order to address legal issues of culpability. It should be possible to apportion culpability among responsible creators (designers and manufacturers) and operators to avoid confusion or fear within the general public".[176]

[170] Ibid.

[171] Ibid. "Principle 5 – 1. For users, what the system is doing and why. 2. For creators, including those undertaking the validation and certification of A/IS, the systems' processes and input data. 3. For an accident investigator, if accidents occur. 4. For those in the legal process, to inform evidence and decision-making. 5. For the public, to build confidence in the technology".

[172] Ibid.

[173] Ibid., Principle 6 (Accountability), at p. 29.

[174] Ibid.

[175] Ibid.

[176] Ibid. The IEEE Report underlined that, in order to best address issues of responsibility and accountability, systems for registration and record-keeping should be established so that it is always possible to find out who is legally responsible for a particular A/IS. Creators, including manufacturers, along with operators, of A/IS should register key, high-level parameters, including: intended use; training data and training environment, if applicable; sensors and real world data sources;

Likewise, the UK House of Lords Artificial Intelligence Committee's Report used the terms 'intelligibility' and 'accountability'. It stressed that the newer generation of deep learning systems (the so-called 'black box' systems) rely on the feeding of information through many different 'layers' of processing in order to produce a particular output or decision. Due to the number and complexity of stages involved in these systems, researchers are not allowed to understand how a prediction/ decision was made; even the developers of these deep learning systems cannot always be certain about the factors that led a system to decide one thing over another.[177] As a result, it highlighted the importance of making AI understandable to developers, users and regulators, it argued that "artificial intelligence should operate on principles of intelligibility and fairness"[178] and that clear principles for accountability and intelligibility need to be established.[179] It pointed out that the terms 'transparency', 'interpretability' and 'explainability' have been used often interchangeably and that, for reasons of simplicity, it opted to use the term 'intelligibility', which covers both technical transparency and explainability.[180]

The UK House of Lords Artificial Intelligence Committee's Report appeared to stress that the degree to which intelligibility is needed is highly dependent on the purposes to which the AI system was to be put and the severity of the consequences if the output/decision of the AI system is erroneous or inaccurate.[181] Some experts argued in favour of restricting the use of unintelligible systems in certain important or safety-critical areas, stressing that, given the "completely unintelligible" nature of decisions made via deep learning systems, it would be more practicable to focus on outcomes, and "only license critical AI systems that satisfy a set of standardised tests, irrespective of the mechanism used by the AI component".[182] On the other hand, other (fewer) witnesses argued that the necessity for intelligible AI systems

algorithms; process graphs; model features, at various levels; user interfaces; actuators and outputs; optimization goals, loss functions, and reward functions. See Principle 6 (Accountability), at p. 30.

[177] The UK House of Lords Artificial Intelligence Committee's Report "AI in the UK: ready, willing and able?", 16 April 2018, at para. 90, p. 36.

[178] Ibid. See the "five overarching principles for an AI code" offered in paragraph 417 of the Report, Principle number (2).

[179] Ibid., at para. 318, p. 98.

[180] Ibid., at para. 91, p. 36.

[181] Ibid., at para. 92, p. 36. See also at para. 94, p. 37. Examples of contexts in which a high degree of intelligibility was thought to be necessary included: (i) Judicial and legal affairs (see written evidence from Joanna Goodman, Dr Paresh Kathrani, Dr Steven Cranfield, Chrissie Lightfoot and Michael Butterworth, Future Intelligence, Ocado Group plc; (ii) Healthcare (see written evidence from Medicines and Healthcare products Regulatory Agency, PHG Foundation, SCAMPI Research Consortium, City, University of London, Professor John Fox); (iii) Certain kinds of financial products and services, for example, personal loans and insurance (see written evidence from Professor Michael Wooldridge); (iv) Autonomous vehicles (see written evidence from Five AI Ltd and UK Computing Research Committee); (v) Weapons systems (see written evidence from Amnesty International and Big Brother Watch).

[182] The UK House of Lords Artificial Intelligence Committee's Report "AI in the UK: ready, willing and able?", 16 April 2018, at para. 94, p. 37. See written evidence from Professor Robert Fisher,

was low since, according to their view: (i) even deep learning systems can meet the intelligibility and auditability demands;[183] (ii) it is unrealistic to hold AI to a higher

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Professor Alan Bundy, Professor Simon King, Professor David Robertson, Dr Michael Rovatsos, Professor Austin Tate and Professor Chris Williams.

According to the Amnesty International [Written evidence (AIC0180), at points 24–27] - Amnesty International United Kingdom Section, a lack of transparency and accountability in current systems denies those who are harmed by AI-informed decisions adequate access to justice or remedy. The inability to scrutinise the workings of all current deep learning systems (the 'black box phenomenon') creates a huge problem with trusting algorithmically-generated decisions. Where AI systems deny someone their rights, understanding the steps taken to deliver that decision is crucial to deliver remedy and justice. Provisions for accountability need to be considered before AI systems become widespread—practically, this may occur at multiple points, including in developing software, using training data responsibly, executing decisions. To what extent will any automated decision be able to be 'overridden', and by whom? Restricting the use of deep learning systems in some cases may be required, where such systems make decisions that directly impact individual rights. The UK government should encourage the development of explainable AI systems, which would be more transparent and allow for effective remedies. Systems need transparency, good governance (including scrutiny of systems and data for potential bias), and accountability measures in place before they are rolled out into public use—especially where AI systems play a decisive and influential role in public services (policing, social care, welfare, state healthcare).

Also, in relation to the question under what conditions a relative lack of transparency in artificial intelligence systems (so-called 'black boxing') is considered to be acceptable (or when it should not be permissible), the Amnesty International argued (points 39–41) that the black box problem is most acute in applications of AI that have the potential to impact on individual rights. Many commercial and non-commercial AI applications may not require the same threshold of accountability, as their impact on individual rights is either remote or negligible. It is vital that AI systems are not rolled out in areas of public life where they could discriminate or generate otherwise unfair decisions without the ability for interrogation and accountability. Where there is potential adverse consequences for individual rights, there must be higher transparency standards applied, with obligations both on the developers of the AI and the institutions using the AI system. This includes: (i) detecting for and correcting for bias in design of the AI and in the data used; (ii) effective mechanisms to guarantee transparency and accountability in use, including regular audits to check for discriminatory decisions and access to remedy when individuals are harmed; (iii) not using AI where there is a risk of harm and no effective means of accountability.

[183] Ibid., at para. 92, pp. 36–37. Nvidia, for example, made the point that the assertion that neural networks lack transparency is false. The nature of neural networks is different from that of hand-written computer programs; therefore, new methods for validation must be, and indeed are being, developed. Machine learning algorithms are often much shorter and simpler than conventional software coding, and are therefore in some respects easier to understand and inspect. In practice, other code validation methods are used (for example unit or coverage tests) to ensure source code is functioning correctly and is secure (security breaches are just lapses in that test methodology). The fact that the code is written in a programming language just provides an illusion of transparency as it is not possible for humans to read and the only validation can be done through thorough tests. From this perspective, neural networks offer significant advantage over the hand-written code. Not only do the standard test procedures still apply (as it is common practice to test them against a test dataset), a neural network's code complexity is negligible (to the extent where the code can be easily read and understood by a single human) and more importantly they can be more formally tested and their performance mathematically proven (since they are just complex functions).

Also, according to Dr Timothy Lanfear, Director of the EMEA Solution Architecture and Engineering team at Nvidia, "We are using systems every day that are of a level of complexity standard than human decision-making, which itself often appears to be "illogical or impenetrable";[184] "equally, one could question whether most human decision making is transparent and highly accurate";[185] (iii) restricting ourselves only to techniques which we can fully understand, can result in limiting what could be accomplished with AI and preventing the development of emerging digital technologies.[186]

According to the UK House of Lords Artificial Intelligence Committee's Report, a solution to the question of intelligibility would be to increase the technical transparency of the AI system, so that experts can be given the opportunity to access the source code of an AI system and thus to understand how an AI system has been built. However, the Report appears to recognize that "this will not necessarily reveal why a particular system made a particular decision in a given situation"; "significantly, it does not include the data that was input into the system in a particular scenario, nor how that data was processed in order to arrive at a decision, and a system may behave very differently with different datasets".[187] As a result, the Report argued that achieving full technical transparency is difficult, and possibly even impossible, for certain kinds of AI systems, and would not be appropriate in a variety of cases. However, technical transparency was considered to be imperative in specific safety-critical areas, in which the relevant regulatory authorities should have the power to intervene and "mandate the use of more transparent forms of AI, even at the potential expense of power and accuracy".[188] The Report also mentioned that explainability,[189] whereby AI systems are built in such a way that they can explain

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that we cannot absorb. Artificial intelligence is no different from that. It is also at a level of complexity that cannot be grasped as a whole. Nevertheless, what you can do is to break this down into pieces, find ways of testing it to check that it is doing the things you expect it to do and, if it is not, take some action".

[184] Ibid., at para. 93, p. 37. See written evidence from Deep Science Ventures; Dr Dan O'Hara, Professor Shaun Lawson, Dr Ben Kirman and Dr Conor Linehan; Professor Robert Fisher, Professor Alan Bundy, Professor Simon King, Professor David Robertson, Dr Michael Rovatsos, Professor Austin Tate and Professor Chris Williams.

[185] Ibid. See written evidence (AIC0029) from Professor Robert Fisher, Professor Alan Bundy, Professor Simon King, Professor David Robertson, Dr Michael Rovatsos, Professor Austin Tate, Professor Chris Williams.

[186] Ibid. See written evidence from Professor Robert Fisher, Professor Alan Bundy, Professor Simon King, Professor David Robertson, Dr Michael Rovatsos, Professor Austin Tate and Professor Chris Williams; Michael Veale; Five AI Ltd; University College London (UCL) and Electronic Frontier Foundation.

[187] Ibid., at para. 95, p. 38.

[188] Ibid., at para. 99, p. 38.

[189] Ibid.

For a distinction between transparency and explainability, see below the responses of Professor Alan Winfield and Dr. Konstantinos Karachalios to Question 22, Select Committee on Artificial Intelligence, Corrected oral evidence: Artificial Intelligence, Tuesday 17 October 2017, Members present: Lord Clement-Jones (The Chairman); Baroness Bakewell, Lord Giddens; Baroness Grender; Lord Hollick; Lord Levene of Portsoken; Viscount Ridley; Baroness Rock; Lord St John of Bletso; Lord Swinfen., Evidence Session No. 3, Heard in Public, Questions 18–28, the information and logic used by machine learning systems to reach a specific decision, could be a more useful approach compared to technical transparency.[190]

Witnesses: Professor Alan Winfield, Professor of Robot Ethics, University of the West of England, Bristol; Dr Ing Konstantinos Karachalios, Managing Director, IEEE Standards Association:

The Chairman: You seem to be making a distinction between transparency and explainability. Would either of you like to unpack that?

Dr Ing Konstantinos Karachalios: Full transparency is having the source code to see how it is written. It is fully transparent but it does not explain anything. You do not understand it.

Professor Alan Winfield: An example might help to clarify. If you have a driverless car and it is involved in an accident, the accident investigators need to understand what happened to cause the accident. That requires transparency, not necessarily explainability. Explainability for the user of a care robot—say it is an elderly person with a care robot in their home—means that that elderly person should be able to ask the robot in some fashion, "Why did you just do that?".

[190] Ibid., at para. 100, p. 39.

It was pointed out that many companies and organisations are working on explanation systems, which would help to consolidate and translate the processes and decisions made by machine learning algorithms into forms that are comprehensible to human operators. Furthermore, some of the largest technology companies, including Google, IBM and Microsoft, mentioned their commitment to developing interpretable machine learning systems.

See, for instance, written evidence from IBM (AIC0160), especially points 28 and 29: "the algorithms that underpin AI systems need to be as transparent, or at least as interpretable as possible. In other words, they need to be able to explain their behaviour in terms that humans can understand — from how they interpreted their input to why they recommended a particular output. To do this, we recommend all AI systems should include "explanation-based collateral systems". The provided explanations should be meaningful to the targeted users. For example, in AI decision support systems whose aim is to help doctors identify the best therapy for a patient, such AI systems need to provide useful explanations to doctors, patients, nurses, relatives, etc. More generally, existing AI systems support many advanced analytical applications for industries like healthcare, financial services and law. In these scenarios, data-centric compliance monitoring and auditing systems can visually explain various decision paths and their associated risks, complete with the reasoning and motivations behind the recommendation. And the parameters for these solutions are defined by existing regulatory requirements specific to that industry".

See also written evidence from Google (AIC0225), point 6.9, where it was mentioned that Glassbox is a machine learning framework optimised for interpretability. It involves creating mathematical models to smooth out the influence of outliers in a data set, thus helping to make results more predictable and decipherable.

Moreover, with regard to Microsoft's development of best practices for intelligible AI systems, see written evidence Microsoft (AIC0149), at point 27: As AI-based systems are increasingly used to make decisions that affect people's lives in important ways, people naturally want to understand how these systems operate, and why they make the recommendations that they do. Enabling transparency of AI systems can be challenging due to their complexity and the fact that recommendations are largely a function of an understanding of massive amounts of data, which computers excel at, but people do not. (And access to data about people often cannot be provided, given privacy considerations.) Researchers are developing a number of promising techniques to help provide transparency, such as developing simpler systems that closely mimic the recommendations of more accurate systems yet are easier to understand, and systems that enable people to vary various inputs to see the effect on system recommendations. Microsoft is working with PAI to develop best practices to enable useful transparency. Such best practices will likely include an explanation of the system objectives, the data sets used to train the system during development and in deployment, selection criteria for the algorithms, the system components and their interactions, testing and validation of the system, and risk mitigation considerations.

According to many witnesses and comments received by the UK House of Lords Artificial Intelligence Committee, Article 22 of the GDPR could decrease the risk and tackle the challenge of explainable AI systems, as it contains a 'right to an explanation' provision which gives individuals, when they have been subject to fully automated decision making (and where the outcome has a significant impact on them), the right to ask for an explanation as to how that decision was reached, or to ask for a human to make the decision.[191] Humans should have the right not to be subject to a decision based solely on automated processing when this significantly affects them or produces legal effects. On the other hand, it was noted that Article 22 of the GDPR does not apply if a decision is based on explicit consent, if it does not produce a legal outcome of similar significant effect on the data subject, or if the process is only semi-automated.[192]

In conclusion, the UK House of Lords Artificial Intelligence Committee's Report pointed out that the crucial question relates to what extend and under which circumstances AI systems should be transparent, and underlined that it should not be acceptable to develop and deploy any AI system which could have a substantial impact on an individual's life, unless it can generate a full and satisfactory explanation for the decisions it will take. A relative lack of transparency in artificial intelligence systems could be acceptable only under specific circumstances, i.e. the degree of acceptable transparency should differ depending on the situation, for

[191] Ibid., at para. 101, p. 39. See written evidence from Future Intelligence (AIC0216); PricewaterhouseCoopers LLP (AIC0162); IBM (AIC0160); CognitionX (AIC0170); Big Brother Watch (AIC0154); Will Crosthwait (AIC0094) and Dr Maria Ioannidou (AIC0082). Also: written evidence from Article 19 (AIC0129); Information Commissioner's Office (AIC0132); CBI (AIC0114) and Ocado Group plc (AIC0050).

[192] Ibid., at para. 101, p. 39.

Article 22 of Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation – GDPR) gives individuals the right not to be subject to a decision based solely on automated processing when this produces legal effects on users or similarly significantly affects them. "Article 22: Automated individual decision-making, including profiling 1. The data subject shall have the right not to be subject to a decision based solely on automated processing, including profiling, which produces legal effects concerning him or her or similarly significantly affects him or her. 2. Paragraph 1 shall not apply if the decision: (a) is necessary for entering into, or performance of, a contract between the data subject and a data controller; (b) is authorised by Union or Member State law to which the controller is subject and which also lays down suitable measures to safeguard the data subject's rights and freedoms and legitimate interests; or (c) is based on the data subject's explicit consent. 3. In the cases referred to in points (a) and (c) of paragraph 2, the data controller shall implement suitable measures to safeguard the data subject's rights and freedoms and legitimate interests, at least the right to obtain human intervention on the part of the controller, to express his or her point of view and to contest the decision. 4. Decisions referred to in paragraph 2 shall not be based on special categories of personal data referred to in Article 9(1), unless point (a) or (g) of Article 9(2) applies and suitable measures to safeguard the data subject's rights and freedoms and legitimate interests are in place".

instance the use of unintelligible systems could be restricted in certain important or safety-critical areas. Companies and organisations should increase the intelligibility of their AI systems and, if this is not done, then regulatory authorities should prohibit the use of opaque and incomprehensible technology in major and sensitive areas of society.[193] As it was stated: "We believe that the development of intelligible AI

[193] The UK House of Lords Artificial Intelligence Committee's Report "AI in the UK: ready, willing and able?", 16 April 2018, at p. 5.

See below the responses of Professor Alan Winfield and Dr. Konstantinos Karachalios to Question 22, Select Committee on Artificial Intelligence, Corrected oral evidence: Artificial Intelligence, Tuesday 17 October 2017, Members present: Lord Clement-Jones (The Chairman); Baroness Bakewell, Lord Giddens; Baroness Grender; Lord Hollick; Lord Levene of Portsoken; Viscount Ridley; Baroness Rock; Lord St John of Bletso; Lord Swinfen., Evidence Session No. 3, Heard in Public, Questions 18–28, Witnesses: Professor Alan Winfield, Professor of Robot Ethics, University of the West of England, Bristol; Dr Ing Konstantinos Karachalios, Managing Director, IEEE Standards Association:

Viscount Ridley: "My question is about transparency. We cannot be far from the point when artificial intelligence diagnoses a disease or offers a legal opinion without being able to explain how it reached that conclusion. That is already true of human beings in some sense, but is it a particular problem with AI, with robots, and how transparent should artificial intelligence systems be?

Professor Alan Winfield: I take a very hard line on this. I think it should always—always—be possible to find out why an AI made a particular decision. That is very easy to say, of course, but we know that in practice it can be very difficult. It seems to me absolutely unacceptable that one might accept the decision of a medical-diagnosis AI or a mortgage application recommender system without understanding why it made that decision. Many members of the AI community will get very cross with me—I am sure they are right now if they are watching—because, of course, what I am effectively saying is that we should not be using systems that are not transparent, such as deep learning systems. I would simply apply an engineering approach whereby we need to be able to understand why the system makes the decisions it does; otherwise, if a system goes wrong and causes harm, we simply cannot find out what went wrong. I would like to offer the analogy of aircraft autopilots. We all understand very well the standards that we set for the engineering of those systems, and an AI autopilot and driverless cars, for instance, should be held to no less a standard of safety, and explainability or transparency.

Viscount Ridley: Is not AlphaGo already failing that threshold?

Professor Alan Winfield: It is, yes.

The Chairman: By which you mean deep learning?

Professor Alan Winfield: My challenge to the AI community—and they are smart guys, smart men and women—is to invent AI systems that are explainable. I do not believe that it is technically impossible.

The Chairman: Has the horse not already bolted, though?

Professor Alan Winfield: It may well have done, except that we can still regulate—and I believe that we should—and say that it is simply not acceptable to have, for instance, a medical-diagnosis AI that cannot be explained.

Viscount Ridley: I have a supplementary question on that that I was supposed to ask. Does the degree of acceptable transparency differ depending on the situation? In other words, some of our evidence has suggested that it will be fine in some circumstances but not in others, so in a game it might be all right and in a medical diagnosis it might not be. Would you accept that?

Professor Alan Winfield: I would indeed. I am focused primarily, as you can probably tell, on safety-critical systems. That is where if the AI goes wrong, physical, financial or psychological harm could result—in other words, where harms result from a failing AI.

The Chairman: Dr Karachalios, do you agree with Professor Winfield's hard line?

systems is a fundamental necessity if AI is to become an integral and trusted tool in our society. Whether this takes the form of technical transparency, explainability, or indeed both, will depend on the context and the stakes involved, but in most cases we believe explainability will be a more useful approach for the citizen and the consumer. This approach is also reflected in new EU and UK legislation. We believe it is not acceptable to deploy any artificial intelligence system which could have a substantial impact on an individual's life, unless it can generate a full and satisfactory explanation for the decisions it will take. In cases such as deep neural networks, where it is not yet possible to generate thorough explanations for the decisions that are made, this may mean delaying their deployment for particular uses until alternative solutions are found".[194]

Likewise, the UNI Global Union Top 10 Principles for Ethical AI used the terms 'transparency' and 'accountability'. In particular, it stated that transparency "is a prerequisite for ascertaining that the remaining principles are observed" and followed the exact definition given by the IEEE Report, namely that transparent is the AI system in which it is possible to discover how and why a particular decision was taken by the said system or, in the case of a robot, why it acted the way it did.[195] It stressed that the opportunity to access the source code of an AI system cannot necessarily present the reason why an AI system took a particular decision: "open source code is neither necessary nor sufficient for transparency – clarity cannot be obfuscated by complexity".[196] It recognised that the trustworthiness of AI systems is a key factor for their diffusion and adoption and, to this end, transparency plays a major role: "for users, transparency is important because it builds trust in, and understanding of, the system, by providing a simple way for the user to understand what the system is doing and why".[197] Again, following the recommendations of the IEEE - Ethically Aligned Design (EAD) Report, it stated that (i) transparency is necessary so that AI systems can be objectively tested, evaluated and validated ("it exposes the system's processes for scrutiny"); (ii) if an accidents occurs, the investigator needs to understand what happened to cause the accident ("so the internal process that led to the accident can be understood") and, therefore, the AI

Dr Ing Konstantinos Karachalios: Yes, and I believe that there are different aspects of it. Transparency is not explainability. It is different. A system that can explain what it does is different from a fully transparent system. There are efforts to make decisions more transparent. There are elements of the European Commission's GDPR that are forcing this transparency. This is the right way to go—and it can be done—to give direction to the researchers and scientists. In addition, it is very difficult to understand why dataset A has been transformed to dataset B, because we do not remember the way computers remember. We do not have this capacity. Even if we see it, we do not understand how it came about. This is a problem that we need to understand, because if you do not get a job or you are refused medical treatment, nobody can tell you why. We need systems that can explain this, I agree with you.

[194] Ibid., at para. 105, p. 40.

[195] The UNI Global Union Top 10 Principles for Ethical Artificial Intelligence, 2017. See "Principle 1 – Demand That AI Systems Are Transparent", at pp. 6–7.

[196] Ibid., "Principle 1 – Demand That AI Systems Are Transparent", (1A), at p. 6.

[197] Ibid., "Principle 1 – Demand That AI Systems Are Transparent", (1B), at p. 6.

systems should be transparent and accountable; (iii) similarly, following an accident, transparency is essential for those involved in the legal process (such as judges, lawyers, expert witnesses, juries) to inform evidence and decision-making; (iv) users affected by an AI system should be enabled to challenge its outcome, to ask for an explanation as to how that decision was reached, or to ask for a human to review it and make the final decision.[198]

Also, in order to best address the issues of transparency and accountability, the UNI Global Union Top 10 Principles for Ethical AI—in line with the proposal of the IEEE's Report ("for accident investigators, secure storage of sensor and internal state data comparable to a flight data recorder or black box")—stressed that AI systems should be equipped with a 'black box' which records data on every transaction carried out by the system, including the rationale that contributed to its decisions: "Full transparency in an AI system should be facilitated by the presence of a device that can record information about said system in the form of an "ethical black box" that not only contains relevant data to ensure transparency and accountability of a system, but also includes clear data and information on the ethical considerations built into said system. Applied to robots, the ethical black box would record all decisions, its bases for decision-making, movements, and sensory data for its robot host. The data provided by the black box could also assist robots in explaining their actions in language human users can understand, fostering better relationships and improving the user experience. The read out of the ethical black box should be uncomplicated and fast".[199]

The AI4People's Scientific Committee, which influenced heavily the HLEG's Ethics Guidelines for Trustworthy AI, noted that "a small fraction of humanity is currently engaged in the design and development of a set of technologies that are already transforming the everyday lives of just about everyone else"[200] and that, at the same time, the programming and decisions of AI systems are not visible and detectable by the general public—the workings of AI "are often invisible or unintelligible to all but (at best) the most expert observers".[201] As a result, in order to build and maintain public trust and confidence in AI systems, it is crucial to understand and provide an explanation as to why an AI system has produced a particular output or decision as well as to ensure accountability, i.e. to be able to identify who should be held liable in the event of failure. Therefore, according to the AI4People's Scientific Committee, the principle of explicability – both in the epistemological sense of "intelligibility" (as an answer to the question "how does it work?") and in the ethical sense of "accountability" (as an answer to the question:

[198] Ibid., "Principle 1 – Demand That AI Systems Are Transparent", (1C) to (1G), at pp. 6–7.
See also Principle 4 – 'Adopt a Human-In-Command Approach', where it is stated that "workers must also have the 'right of explanation' when AI systems are used in human-resource procedures, such as recruitment, promotion or dismissal", at p. 8.

[199] Ibid., "Principle 2 – Equip AI Systems With an "Ethical Black Box"", at p. 7.

[200] Floridi et al. (2018), p. 699.

[201] Ibid., at p. 700.

"who is responsible for the way it works?")—is "the crucial missing piece of the jigsaw when we seek to apply the framework of bioethics to the ethics of AI".[202] It concluded by arguing that the principle of explicability enables and complements the other principles through intelligibility and accountability: "for AI to be beneficent and non-maleficent, we must be able to understand the good or harm it is actually doing to society, and in which ways; for AI to promote and not constrain human autonomy, our 'decision about who should decide' must be informed by knowledge of how AI would act instead of us; and for AI to be just, we must ensure that the technology – or, more accurately, the people and organisations developing and deploying it – are held accountable in the event of a negative outcome, which would require in turn some understanding of why this outcome arose".[203]

(v) The Principle of Beneficence ('Do Only Good')

It should be stressed that the AI4People's Scientific Committee—alongside the four ethical principles contained in the HLEG's Ethics Guidelines for Trustworthy AI—cited the 'principle of beneficence' ("promoting well-being, preserving dignity, and sustaining the planet") as the principle of creating AI technology that is beneficial to humanity: "the prominence of beneficence firmly underlines the central importance of promoting the well-being of people and the planet with AI".[204]

The principle of beneficence was not explicitly cited in the revised HLEG's Ethics Guidelines for Trustworthy AI.[205] However, it was mentioned as one of the seven requirements to achieve Trustworthy AI ('societal and environmental well-being'), where it is stated that the protection of human dignity should be ensured[206] and that the prevention of harm principle includes consideration of the natural environment and all living beings;[207] "in line with the principles of fairness and prevention of harm, the broader society, other sentient beings and the environment should be also considered as stakeholders throughout the AI system's life cycle.

[202] Ibid.

[203] Ibid.

[204] Ibid., at pp. 696–697. See also: "A Unified Framework of Five Principles for AI in Society", by Luciano Floridi and Josh Cowls, Harvard Data Science Review, July 2019 (updated in November 2019), at p. 6.

[205] See, for instance, the Document "Ethics Guidelines for Trustworthy AI – Overview of the main comments received through the Open Consultation", at p. 4: "The inclusion of the principle "do good" received a number of critical comments, as this was not found to be a principle that could be a moral imperative in each and every case (e.g. when pursuing fundamental research) and not well suited in the context of AI. A few contributors suggested the addition of other principles. The revised Guidelines no longer contain a reference to the "do good" principle. However, they stipulate clearly that one of the goals of Trustworthy AI is to improve individual and collective wellbeing, and provide guidance concerning how ethics can help to achieve this objective".

[206] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at p. 12.

[207] Ibid.

Sustainability and ecological responsibility of AI systems should be encouraged, and research should be fostered into AI solutions addressing areas of global concern, such as for instance the Sustainable Development Goals. Ideally, AI systems should be used to benefit all human beings, including future generations";[208] "for AI to be trustworthy, its impact on the environment and other sentient beings should be taken into account. Ideally, all humans, including future generations, should benefit from biodiversity and a habitable environment. Sustainability and ecological responsibility of AI systems should hence be encouraged. The same applies to AI solutions addressing areas of global concern, such as for instance the UN Sustainable Development Goals. Furthermore, the impact of AI systems should be considered not only from an individual perspective, but also from the perspective of society as a whole. The use of AI systems should be given careful consideration particularly in situations relating to the democratic process, including opinion-formation, political decision-making or electoral contexts. Moreover, AI's social impact should be considered".[209]

In the EGE's Statement on AI, the principle of beneficence is cited under the principles titled 'human dignity' ("the principle of human dignity, understood as the recognition of the inherent human state of being worthy of respect, must not be violated by 'autonomous' technologies")[210] and "sustainability" ("AI technology must be in line with the human responsibility to ensure the basic preconditions for life on our planet, continued prospering for mankind and preservation of a good environment for future generations. Strategies to prevent future technologies from detrimentally affecting human life and nature are to be based on policies that ensure the priority of environmental protection and sustainability").[211] Moreover, the EGE's Statement on AI, under the principle titled "justice, equity, and solidarity", emphasized the importance of justice and stressed that "AI should contribute to global justice and equal access to the benefits and advantages that AI, robotics and 'autonomous' systems can bring".[212]

The importance of the principle of beneficence is illustrated in the Montreal Declaration for a responsible development of AI, which states that the aim should be "the transition towards a society in which AI helps promote the common good: algorithmic governance, digital literacy, digital inclusion of diversity and ecological sustainability".[213] This "common good" could be promoted by AI "which can

[208] Ibid., at p. 19.

[209] Commission Communication, "Building Trust in Human-Centric Artificial Intelligence", COM (2019) 168 final, Brussels, 8.4.2019, at p. 6.

[210] European Group on Ethics in Science and New Technologies (EGE), "Statement on Artificial Intelligence, Robotics and 'Autonomous' Systems", Brussels, 9 March 2018, European Commission, Directorate-General for Research and Innovation, at p. 16.

[211] Ibid., at p. 19.

[212] Ibid., at p. 17.

[213] One of the objectives of the Montreal Declaration for a Responsible Development of AI, alongside the development of an ethical framework for the deployment of AI so that everyone benefits from this technological revolution, is "to collectively achieve equitable, inclusive, and

generate considerable social benefits by improving living conditions and health, facilitating justice, creating wealth, bolstering public safety, and mitigating the impact of human activities on the environment and the climate".[214] In relation to the beneficence principle, the Montreal Declaration for a responsible development of AI uses the term 'well-being principle' and states that "the development and use of artificial intelligence systems (AIS) must permit the growth of the well-being of all sentient beings"[215] and that "AIS must help individuals improve their living conditions, their health, and their working conditions".[216] Moreover, it highlighted that "the AIS must allow individuals to fulfill their own moral objectives and their conception of a life worth living",[217] and emphasized, under the 'sustainable development principle', that "the development and use of AIS must be carried out so as to ensure a strong environmental sustainability of the planet".[218]

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ecologically sustainable AI development". See "The Montreal Declaration for a Responsible Development of Artificial Intelligence", December 2018, at p. 5.

[214] Ibid., at p. 7.

[215] Ibid., Principle number 1 ("Well-Being Principle"), at p. 8.

[216] Ibid., Principle number 1 ("Well-Being Principle"), 1.1, at p. 8.

> 1 – Well-Being Principle: The development and use of artificial intelligence systems (AIS) must permit the growth of the well-being of all sentient beings. 1. AIS must help individuals improve their living conditions, their health, and their working conditions. 2. AIS must allow individuals to pursue their preferences, so long as they do not cause harm to other sentient beings. 3. AIS must allow people to exercise their mental and physical capacities. 4. AIS must not become a source of ill-being, unless it allows us to achieve a superior well-being than what one could attain otherwise. 5. AIS use should not contribute to increasing stress, anxiety, or a sense of being harassed by one's digital environment.

[217] Ibid., Principle number 2 ("Respect for Autonomy Principle"), 2.1, at p. 9.

[218] Ibid., Principle number 10 ("Sustainable Development Principle"), at p. 17.

> 10 – Sustainable Development Principle: The development and use of AIS must be carried out so as to ensure a strong environmental sustainability of the planet. 1. AIS hardware, its digital infrastructure and the relevant objects on which it relies such as data centers, must aim for the greatest energy efficiency and to mitigate greenhouse gas emissions over its entire life cycle. 2. AIS hardware, its digital infrastructure and the relevant objects on which it relies, must aim to generate the least amount of electric and electronic waste and to provide for maintenance, repair, and recycling procedures according to the principles of circular economy. 3. AIS hardware, its digital infrastructure and the relevant objects on which it relies, must minimize our impact on ecosystems and biodiversity at every stage of its life cycle, notably with respect to the extraction of resources and the ultimate disposal of the equipment when it has reached the end of its useful life. 4. Public and private actors must support the environmentally responsible development of AIS in order to combat the waste of natural resources and produced goods, build sustainable supply chains and trade, and reduce global pollution.

According to the Montreal Declaration for a Responsible Development of AI, the terms strong environmental sustainability and sustainable development are explained as follows: (i) Strong Environmental Sustainability: the notion of strong environmental sustainability goes back to the idea that in order to be sustainable, the rate of natural resource consumption and polluting emissions must be compatible with planetary environmental limits, the rate of resources and ecosystem

Likewise, the Partnership on AI (PAI)—Tenets stressed that AI should be developed for the benefit of humanity ("AI technologies hold great promise for raising the quality of people's lives and can be leveraged to help humanity address important global challenges such as climate change, food, inequality, health, and education"[219] and for the common good ('AI and Social Good': "AI offers great potential for promoting the public good, for example in the realms of education, housing, public health, and sustainability"). [220] To this end, it pointed out that one of its main goals is to "identify and foster aspirational efforts in AI for socially beneficial purposes".[221] The aim of serving the social good is also illustrated when it is stated that "we will seek to ensure that AI technologies benefit and empower as many people as possible";[222] "we will work to maximize the benefits . . . of AI technologies, by: . . . b) striving to understand and respect the interests of all parties that may be impacted by AI advances; c) working to ensure that AI research and engineering communities remain socially responsible, sensitive, and engaged directly with the potential influences of AI technologies on wider society".[223]

Similarly, the Asilomar AI Principles document characterized the beneficence principle as "common good", "shared prosperity", "beneficial intelligence and beneficial use", and stated that AI should be developed for the benefit of all humanity. In particular, it stressed that "the goal of AI research should be to create not undirected intelligence, but beneficial intelligence",[224] that "investments in AI should be accompanied by funding for research on ensuring its beneficial use"[225] and that we need to answer questions as to how we could "grow our prosperity through automation while maintaining people's resources and purpose".[226] The Asilomar AI Principles document underlined the need for "shared benefit" and "shared prosperity" stemming from AI. It stated that AI development should produce benefits for all ("Shared Benefit - AI technologies should benefit and empower as many people as possible")[227] and that these benefits should be equally and fairly distributed: "Shared Prosperity - The economic prosperity created by AI should be

renewal, and climate stability. Unlike weak sustainability, which requires less effort, strong sustainability does not allow the substitution of the loss of natural resources with artificial capital. (ii) Sustainable Development: sustainable development refers to the development of human society that is compatible with the capacity of natural systems to offer the necessary resources and services to this society. It is economic and social development that fulfills current needs without compromising the existence of future generations (at p. 20 of the Declaration).

219 "Partnership on AI (PAI) – Tenets", 2018, Preamble.

220 Ibid., Thematic Pillar 6.

221 Ibid., Our Goals.

222 Ibid., Tenet number 1.

223 Ibid., Tenet number 6(b) and 6(c).

224 The "Asilomar AI Principles" document, 2017, developed under the auspices of the Future of Life Institute. See Principle number 1 ("Research Goal").

225 Ibid., Principle number 2 ("Research Funding").

226 Ibid., Principle number 2 ("Research Funding").

227 Ibid., See Principle number 14 ("Shared Benefit").

shared broadly, to benefit all of humanity".[228] And, under the "Common Good" principle (listed in the 'longer-term issues' category), it underlined that "superintelligence should only be developed in the service of widely shared ethical ideals, and for the benefit of all humanity rather than one state or organization".[229]

Equally, the UK House of Lords Artificial Intelligence Committee's Report stated that AI should serve humanity and promote the common good. According to two of the five primary principles proposed for an AI Code, "Artificial intelligence should be developed for the common good and benefit of humanity",[230] and all citizens should be able "to flourish mentally, emotionally and economically alongside artificial intelligence". [231]

The UNI Global Union Top 10 Principles for Ethical AI stated that all should have access to the benefits provided by AI development and that those benefits should be fairly shared: "AI technologies should benefit and empower as many people as possible. The economic prosperity created by AI should be distributed broadly and equally, to benefit all of humanity."[232] The beneficence principle is cited as part of the principle titled 'Make AI serve people and planet' which states, *inter alia*, that AI systems must protect and even improve our planet's ecosystems and biodiversity".[233]

The OECD's Recommendation of the Council on AI identified the benefits brought by AI technologies, which have "the potential to improve the welfare and well-being of people, to contribute to positive sustainable global economic activity, to increase innovation and productivity, and to help respond to key global challenges".[234] To this end, the OECD's Recommendation stated, under the principle titled 'inclusive growth, sustainable development and well-being', that "stakeholders should proactively engage in responsible stewardship of trustworthy AI in pursuit of beneficial outcomes for people and the planet, such as augmenting human capabilities and enhancing creativity, advancing inclusion of underrepresented populations, reducing economic, social, gender and other inequalities, and protecting natural environments, thus invigorating inclusive growth, sustainable development and well-being".[235]

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[228] Ibid., Principle number 15 ("Shared Prosperity").

[229] Ibid., Principle number 23 ("Common Good").

[230] The UK House of Lords Artificial Intelligence Committee's Report "AI in the UK: ready, willing and able?", published on 16 April 2018. See the "five overarching principles for an AI code" offered in paragraph 417 of the Report, p. 125, Principle number (1).

[231] Ibid., see the "five overarching principles for an AI code" offered in paragraph 417 of the Report, p. 125, Principle number (4).

[232] The UNI Global Union Top 10 Principles for Ethical Artificial Intelligence, 2017. See Principle 6 - Share the Benefits of AI Systems, at p. 8.

[233] Ibid., at p. 7.

[234] OECD, Recommendation of the Council on Artificial Intelligence, 2020, OECD/LEGAL/0449, adopted on 22.05.2019, at p. 6.

[235] Ibid., at p. 7. Principle 1.1. – Inclusive growth, sustainable development and well-being.

The beneficence principle (and the relevant notions/terms of 'human well-being', 'common good', 'sustainable development for all', 'shared prosperity', 'serving humanity') is celebrated and found throughout the IEEE - Ethically Aligned Design (EAD) Report. First, it is contained in the sub-title of the Report: "A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems (A/IS)". Moreover, it is stated that the ethical general principles created by the IEEE Report "embody the highest ideals of human beneficence within human rights" and "prioritize benefits to humanity and the natural environment from the use of A/IS over commercial and other considerations. Benefits to humanity and the natural environment should not be at odds—the former depends on the latter. Prioritizing human well-being does not mean degrading the environment".[236] More specifically, the IEEE Report stated that autonomous and intelligent systems (A/IS) "must be developed and should operate in a way that is beneficial to people and the environment, beyond simply reaching functional goals and addressing technical problems";[237] "ultimately, our goal should be eudaimonia, a practice elucidated by Aristotle that defines human well-being, both at the individual and collective level, as the highest virtue for a society. Translated roughly as "flourishing", the benefits of eudaimonia begin with conscious contemplation, where ethical considerations help us define how we wish to live";[238] "autonomous and intelligent systems should prioritize and have as their goal the explicit honoring of our inalienable fundamental rights and dignity as well as the increase of human flourishing and environmental sustainability".[239]

The principle of "Well-Being"[240] is cited amongst the eight general principles proposed by the IEEE Report: "A/IS creators shall adopt increased human well-

[236] The IEEE (Institute of Electrical and Electronics Engineers) Global Initiative on Ethics of Autonomous and Intelligent Systems (A/IS) - Ethically Aligned Design (EAD) Report: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems (A/IS), 2019, at p. 17.

[237] Ibid., at p. 2.

[238] Ibid.

[239] Ibid.

[240] Ibid., at pp. 70–71. According to the IEEE Report, and for the purposes of the Ethically Aligned Design, "the term 'well-being' refers to an evaluation of the general quality of life of an individual and the state of external circumstances. The conception of well-being encompasses the full spectrum of personal, social, and environmental factors that enhance human life and on which human life depend. The concept of well-being shall be considered distinct from moral or legal evaluation" (at p. 70). The IEEE Study (at p. 71) argued that the most recognized aspects of well-being are (in alphabetical order) the following: • Community: Belonging, Crime & Safety, Discrimination & Inclusion, Participation, Social Support • Culture: Identity, Values • Economy: Economic Policy, Equality & Environment, Innovation, Jobs, Sustainable Natural Resources & Consumption & Production, Standard of Living • Education: Formal Education, Lifelong Learning, Teacher Training • Environment: Air, Biodiversity, Climate Change, Soil, Water • Government: Confidence, Engagement, Human Rights, Institutions • Human Settlements: Energy, Food, Housing, Information & Communication Technology, Transportation • Physical Health: Health Status, Risk Factors, Service Coverage • Psychological Health: Affect (feelings), Flourishing, Mental Illness & Health, Satisfaction with Life • Work: Governance, Time Balance, Workplace Environment.

being as a primary success criterion for development";[241] "A/IS should prioritize human well-being as an outcome in all system designs, using the best available and widely accepted well-being metrics as their reference point";[242] "A/IS technologies can be narrowly conceived from an ethical standpoint. They can be legal, profitable, and safe in their usage, yet not positively contribute to human and environmental well-being. This means technologies created with the best intentions, but without considering well-being, can still have dramatic negative consequences on people's mental health, emotions, sense of themselves, their autonomy, their ability to achieve their goals, and other dimensions of well-being".[243] Furthermore, in Section 1 of the IEEE Report, under the title "A/IS in Service to Sustainable Development for All", it is underlined that AI technology should serve humanity by improving the quality of life and spreading equally the benefits brought by AI technologies (contributing to shared prosperity), in order to drive the well-being of all people, i.e. ensuring that AI systems are developed and deployed "for the common good". As the IEEE—Ethically Aligned Design (EAD) Report stated: "A/IS could also leverage quality and better standards of life and protect people's dignity, while maintaining cultural diversity and protecting the environment";[244] "the ethical imperative driving this chapter is that A/IS must be harnessed to benefit humanity, promote equality, and realize the world community's vision of a sustainable future";[245] "the inequality in accessing and using the internet, both within and among countries, raises questions on how to spread A/IS benefits across humanity. Ensuring A/IS 'for the common good' is an ethical imperative and at the core of Ethically Aligned Design, First Edition; the key elements of this "common good" are that it is human-centered, accountable, and ensure outcomes that are fair and inclusive";[246] "this chapter explores the imperative for A/IS to serve humanity by improving the quality and standard of life for all people everywhere".[247]

[241] Ibid., at pp. 11, 18, 21–22.

[242] Ibid., at p. 21.

For further reading on this topic, see: IEEE P7010™, Well-being Metric for Autonomous and Intelligent Systems; OECD Guidelines on Measuring Subjective Well-being, 2013; OECD Better Life Index, 2017; United Nations Sustainable Development Goal (SDG) Indicators, 2018.

[243] Ibid., at p. 21.

[244] Ibid., at p. 140.

[245] Ibid.

[246] Ibid., at p. 141.

[247] Ibid.

3.7.4 Tensions Between the Principles and Trade-Offs: Impact-Based Approach

The HLEG's Ethics Guidelines for Trustworthy AI recognised that tensions may arise between the four ethical principles, for which there is no fixed solution. For instance, there might be cases where the principle of prevention of harm may conflict with the principle of human autonomy ("consider as an example the use of AI systems for 'predictive policing', which may help to reduce crime, but in ways that entail surveillance activities that impinge on individual liberty and privacy").[248] Moreover, it was stated that the benefits derived from AI systems should substantially exceed the anticipated individual risks and, therefore, adequate measures should be taken in order to mitigate these risks and ensure that the selected measures are compatible with the principle of proportionality.[249]

It was underlined that special attention should be paid to vulnerable persons and groups, such as the elderly, children, ethnic minorities, people with disabilities and others that have historically been disadvantaged or are at risk of exclusion, as well as to situations which are characterised by asymmetries of power or information, such as between employers and employees, businesses and consumers or governments and citizens.[250] It was argued that AI practitioners should always examine the options available and proceed with an assessment of the anticipated benefits and potential costs of the selected measure (i.e. costs which a measure may entail against its prospective benefits). In order to address the potential tensions between the ethical principles, the decision of AI practitioners should incorporate a discussion on the proportionality of the measure selected and include consideration of alternative measures where possible so that the least burdensome measure can be implemented.

Therefore, tensions between ethical principles would lead to inevitable trade-offs, following an impact-based approach. Such trade-offs should be reasoned, properly documented and explicitly acknowledged and evaluated in terms of their risk to fundamental rights, i.e. preference should be given to the one that is least adverse to fundamental rights. As it was stated, "methods of accountable deliberation to deal with such tensions should be established" and AI practitioners "should approach ethical dilemmas and trade-offs via reasoned, evidence-based reflection rather than intuition or random discretion".[251] At the same time, however, it was stated that, in

[248] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at p. 13.

[249] Ibid. See also at p. 14: "Acknowledge that, while bringing substantial benefits to individuals and society, AI systems also pose certain risks and may have a negative impact, including impacts which may be difficult to anticipate, identify or measure (e.g. on democracy, the rule of law and distributive justice, or on the human mind itself.) Adopt adequate measures to mitigate these risks when appropriate, and proportionately to the magnitude of the risk".

[250] "Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019, at p. 13.

[251] Ibid.

situations in which no ethically acceptable trade-offs can be identified, the development, distribution and use of the AI system in question should not proceed in that form, as “certain fundamental rights and correlated principles are absolute and cannot be subject to a balancing exercise (e.g. human dignity)”.[252]

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[252] Ibid.

Some of the fundamental rights of the ‘Charter of Fundamental Rights of the European Union’ (‘the Charter’ - OJ C 83, 30.03.2010, p0. 389–403) are guaranteed in absolute terms, which means that they cannot be subject to ‘limitations’ or ‘restrictions’. Measures taken by public authorities that interfere with a right protected in absolute terms amount to a violation (an infringement) of this fundamental right. There are only a small number of fundamental rights which are guaranteed in absolute terms. The Charter itself does not explicitly list those rights, but in the light of the case law of the European Courts it is argued that the prohibition of torture and inhuman or degrading treatment or punishment (Article 4 of the Charter) and the prohibition of slavery or servitude (Article 5 of the Charter) are protected in absolute terms. For instance, the prohibition of torture and inhuman or degrading treatment or punishment as enshrined in Article 4 of the Charter is absolute. It is, therefore, not possible to ‘balance’ this prohibition against interests of national security. See settled case law of the European Court of Human Rights, e.g. ECtHR (Grand Chamber), Saadi v. Italy, Application no. 37201/06, Judgment of 28 February 2008, at para. 140.

On the other hand, other fundamental rights could be limited and, therefore, measures taken by public authorities that interfere with such rights could be justified under certain conditions. The interference will only amount to a violation of such a right when the justifying conditions cannot be fulfilled. The requirements for a justified limitation are set out in Article 52 of the Charter, which reads as follows: “Any limitation on the exercise of the rights and freedoms recognised by this Charter must be provided for by law and respect the essence of those rights and freedoms. Subject to the principle of proportionality, limitations may be made only if they are necessary and genuinely meet objectives of general interest recognised by the Union or the need to protect the rights and freedoms of others”. When applying Article 52 of the Charter, the following questions should be addressed: (a) Are the limitations provided for by law and formulated in a clear and predictable manner? (b) Are the limitations necessary to achieve an objective of general interest recognised by the Union or to protect the rights and freedoms of others (which)? (c) Are the limitations proportionate, i.e. appropriate for attaining the objective pursued and not going beyond what is necessary to achieve it? Is there an equally effective but less intrusive measure available? (d) Do the limitations preserve the essence of the fundamental rights concerned?

The Charter does not explicitly define the term ‘limitation’ in the sense of Article 52. In general, however, any measure or conduct by public authorities, such as legislative acts, administrative decisions or state practice, which directly or indirectly affect in a negative way the exercise or enjoyment of the rights and freedoms guaranteed by the Charter, can be considered a ‘limitation’. As an example, the collection, use, or even mere storing [see the European Court of Human Rights, S. and Marper v. UK, Applications nos. 30562/04 and 30566/04, Judgment of 4 December 2008 at para. 67] of information by public authorities about an individual is a limitation of the right to protection of personal data, guaranteed under Article 8 of the Charter, which requires justification. For instance, an official census which includes compulsory questions relating to the individual’s sex, marital status, place of birth and other personal details; the recording of fingerprinting, photography and other personal information by the police (even if the police register is secret); the collection of medical data and the maintenance of medical records; the compulsion by state authorities to reveal details of personal expenditure; records relating to past criminal cases; information relating to terrorist activity, or collecting personal information in order to protect national security.

See also Commission’s Staff Working Paper “Operational Guidance on taking account of Fundamental Rights in Commission Impact Assessments”, Brussels, 6.5.2011, SEC(2011) 567 final, at pp. 9–10.

Legal, Policy Documents and Other Sources

Report of the Committee on Legal Affairs of the European Parliament with recommendations to the Commission on Civil Law Rules on Robotics – Motion for a European Parliament Resolution (2015/2103(INL)), 27.1.2017, A8-0005/2017.

European Parliament Resolution of 16 February 2017 with recommendations to the Commission on Civil Law Rules on Robotics (2015/2103(INL)), (2018/C 252/25), OJ C 252, 18.7.2018, pp. 239–257 (the "European Parliament Resolution of 16 February 2017").

General Secretariat of the Council, Subject: European Council meeting (19 October 2017) – Conclusions, Brussels, 19 October 2017 (OR. en), EUCO 14/17.

Commission Communication "Artificial Intelligence for Europe", COM(2018) 237 final, Brussels, 25.4.2018.

Commission Communication "Coordinated Plan on Artificial Intelligence", COM (2018) 795 final, Brussels, 7.12.2018.

Annex to the Commission Communication 'Coordinated Plan on Artificial Intelligence', COM(2018) 795 final, Brussels, 7.12.2018 - Coordinated Plan on the Development and Use of Artificial Intelligence Made in Europe – 2018.

"Draft Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 18 December 2018.

"Ethics Guidelines for Trustworthy AI", High-Level Expert Group on Artificial Intelligence, 8 April 2019 (the "Ethics Guidelines for Trustworthy AI, 2019").

"A definition of AI: Main capabilities and scientific disciplines – Definition developed for the purpose of the deliverables of the High-Level Expert Group on AI", High-Level Expert Group on Artificial Intelligence, 18 December 2018.

Commission Communication, "Building Trust in Human-Centric Artificial Intelligence", COM(2019) 168 final, Brussels, 8.4.2019.

Commission Communication on "Enabling the digital transformation of health and care in the Digital Single Market; empowering citizens and building a healthier society", Brussels, 25.4.2018, COM(2018) 233 final.

Opinion of the Data Ethics Commission (Datenethikkommission), Publisher: Data Ethics Commission of the Federal Government, Berlin, December 2019, pp. 12–227.

"Human Rights in the Age of Artificial Intelligence", Report by Access Now, November 2018.

"EU backs AI regulation while China and US favour technology", Siddharth Venkataramakrishnan, Financial Times, 25.04.2019.

European Group on Ethics in Science and New Technologies (EGE), "Statement on Artificial Intelligence, Robotics and 'Autonomous' Systems", Brussels, 9 March 2018, European Commission, Directorate-General for Research and Innovation ("EGE – Statement on AI, 2018").

IEEE's (Institute of Electrical and Electronics Engineers) policy paper on 'Ethically Aligned Design', 2016.

ITU's (International Telecommunication Union) Global Summit 'AI for Good', 2017.
ACM's (Association for Computing Machinery) – AAAI/ACM 'Conference on AI, Ethics, and Society', February 2018.
European Parliament, Committee on Legal Affairs 2015/2103 (INL) Report with Recommendations to the Commission on Civil Law Rules on Robotics, Rapporteur Mady Delvaux.
"A Unified Framework of Five Principles for AI in Society", by Luciano Floridi and Josh Cowls, Harvard Data Science Review, July 2019 (updated in November 2019).
"The Montreal Declaration for a Responsible Development of Artificial Intelligence", developed under the auspices of the University of Montreal, following the Forum on the Socially Responsible Development of AI of November 2017. Final text, December 2018.
The IEEE Initiative on Ethics of Autonomous and Intelligent Systems - Ethically Aligned Design (EAD), 2019.
The UK House of Lords Artificial Intelligence Committee's Report "AI in the UK: ready, willing and able?", 16 April 2018.
"Partnership on AI (PAI) – Tenets of the Partnership on AI to Benefit People and Society", 2018. Available at: https://partnershiponai.org/about/#tenets
"Asilomar AI Principles", developed under the auspices of the Future of Life Institute, 2017. Available at: https://futureoflife.org/ai-principles/
UNESCO, 'Report of COMEST on Robotics Ethics', World Commission on the Ethics of Scientific Knowledge and Technology, 2017.
Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation – GDPR).
OECD, Recommendation of the Council on Artificial Intelligence, 2020, OECD/LEGAL/0449, adopted on 22.05.2019.
The UNI Global Union Top 10 Principles for Ethical Artificial Intelligence, 2017.
The IEEE (Institute of Electrical and Electronics Engineers) Global Initiative on Ethics of Autonomous and Intelligent Systems (A/IS) - Ethically Aligned Design (EAD) Report: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems (A/IS), 2019 ("IEEE – EAD Report").
Case C-331/88, *Fedesa and others* [1990] ECR I-4023.
Case C-210/03, judgment of the Court of 14.12.2004.
Case 137/85 *Maizena* [1987] ECR 4587.
Case C-339/92 *ADM Olmuhlen* [1993] ECR I-6473.
Case C-210/00 *Kaserei Champignon Hofmeister* [2002] ECR I-6453.
OECD Guidelines on Measuring Subjective Well-being, 2013.
OECD Better Life Index, 2017.
United Nations Sustainable Development Goal (SDG) Indicators, 2018.
ECtHR (Grand Chamber), Saadi v. Italy, Application no. 37201/06, Judgment of 28 February 2008 at para. 140.

European Court of Human Rights, S. and Marper v. UK, Applications nos. 30562/04 and 30566/04, Judgment of 4 December 2008.

Commission's Staff Working Paper "Operational Guidance on taking account of Fundamental Rights in Commission Impact Assessments", Brussels, 6.5.2011, SEC(2011) 567 final.

References

Brandom R (2018) Amazon's facial recognition matched 28 members of Congress to criminal mugshots. The Verge, July 26, 2018

Carpenter J (2015) Google's algorithm shows prestigious job ads to men, but not to women. Here's why that should worry you. The Washington Post, July 6, 2015

Cath C (2018) Governing artificial intelligence: ethical, legal and technical opportunities and challenges. Philos Transact R Soc A 376:2133

Chopra S, White LF (2011) A legal theory for autonomous artificial agents. The University of Michigan Press, Ann Arbor

Dautenhahn K, Ogden B, Quick T (2002) From embodied to socially embedded agents—implications for interaction-aware robots. Cogn Syst Res 3(3):397–428

Di Nucci E (2017) Sexual rights, disability, and sex robots. In: Danaher J, McArthur N (eds) Robot sex: social and ethical implications. MIT Press, Cambridge, pp 73–88

Dicke K (2002) The founding function of human dignity in the universal declaration of human rights. In: Kretzmer D, Klein E (eds) Concept of human dignity in human rights discourse. Columbia University Press, New York, pp 111–120

Ebers M, Navas Navarro S (eds) (2020) Regulating AI and robotics. In: Algorithms and law. Cambridge University Press, Cambridge, pp 37–99

Fiske A, Henningsen P, Buyx A (2019) Your robot therapist will see you now: Ethical implications of embodied artificial intelligence in psychiatry, psychology, and psychotherapy. J Med Int Res 21(5):e13216

Floridi L (2013) The ethics of information. Oxford University Press, Oxford

L. Floridi (2014) The fourth revolution. How the infosphere is reshaping human reality. Oxford University Press, Oxford

Floridi L (2016) On human dignity as a foundation for the right to privacy. Philos Technol 29(4): 307–312

Floridi L (2018) Soft ethics and the governance of the digital. Philos Technol 31(1):1–8

Floridi L, Cowls J, Beltrametti M, Chatila R, Chazerand P, Dignum V, Luetge C, Madelin R, Pagallo U, Rossi F, Schafer B, Valcke P, Vayena EJM (2018) AI4People – an ethical framework for a good AI society: opportunities, risks, principles, and recommendations. Minds Mach 28(4):689–707

Fosch-Villaronga E (2019) Robots, healthcare, and the law: regulating automation in personal care. Routledge

Fosch-Villaronga E, Albo-Canals J (2019) 'I'll take care of you', said the robot. Paladyn. J Behav Robot 10(1):77–93

Fosch-Villaronga E, Golia A (2019) Robots, standards and the law: Rivalries between private standards and public policymaking for robot governance. Comput Law Secur Rev 35(2): 129–144

Fosch-Villaronga E, Heldeweg M (2018) "'Regulation, I presume?" said the robot—Towards an iterative regulatory process for robot governance. Comput Law Secur Rev 34(6):1258–1277

Fosch-Villaronga E, Millard C (2019) Cloud robotics law and regulation: challenges in the Governance of complex and dynamic cyber-physical ecosystems. Robot Auton Syst 119:77–91

Graeme L, Harmon SHE, Arzuaga F (2012) Foresighting futures: law, new technologies and the challenges of regulating for uncertainty. Law Innov Technol 4(1):1–33
Grodzinsky F, Miller KW, Wolf MJ (2008) The ethics of designing artificial agents. Ethics Inf Technol 10(2-3):115–121
Guihot M, Matthew AF, Suzor NP (2017) Nudging robots: innovative solutions to regulate artificial intelligence. Vanderbilt J Entertain Technol Law 20:385
Hilgendorf E (2018) Problem areas in the dignity debate and the ensemble theory of human dignity. In: Grimm D, Kemmerer A, Möllers C (eds) Human dignity in context. Explorations of a contested concept. Nomos Verlagsgesellschaft mbH & Co. KG, Baden-Baden, p. 325
Koops BJ, Leenes R (2014) Privacy regulation cannot be hardcoded. A critical comment on the 'privacy by design' provision in data-protection law. Int Rev Law Comput Technol 28(2): 159–171
Kritikos M (2016) Legal and ethical reflections concerning robotics. STOA Policy Briefing, June 2016 – PE 563.501, Brussels, European Parliament Research Service
Marchant GE, Allenby BR, Herkert JR (2011) The growing gap between emerging technologies and legal-ethical oversight: the pacing problem, vol 7. Springer Science & Business Media, Berlin
McCrudden C (2008) Human dignity and judicial interpretation of human rights. Eur J Int Law (EJIL) 19(4):655–724
Neuman GL (2000) Human dignity in United States constitutional law. In: Simon D, Weiss M (eds) Zur Autonomie des Individuums. Liber Amicorum Spiros Simitis, Baden-Baden, pp 249–271
O'Mahony C (2012) There is no such thing as a right to dignity. Int J Const Law 10(2):551–574
Schroeder D, Bani-Sadr AH (2017) Dignity in the 21st century: middle east and west. Springer, Berlin
Sharkey A (2014) Robots and human dignity: a consideration of the effects of robot care on the dignity of older people. Ethics Inf Technol 16(1):63–75
Sweeney L (2013) Discrimination in online ad delivery. Harvard University, January 28, 2013
Tzimas T (2021) Legal and ethical challenges of artificial intelligence from an international law perspective. Springer, Cham, Switzerland, pp 9–32
Veale M, Binns R, Edwards L (2018) Algorithms that remember: model inversion attacks and data protection law. Philos Transact R Soc A Math Phys Eng Sci 376(2133):20180083
Wagner B (2018) Ethics as an escape from regulation: from ethics-washing to ethics-shopping? In: Hildebrandt M (ed) Being profiling. Cogitas ergo sum. Amsterdam University Press, Amsterdam, pp 84–90
Weisstub DN (2002) Honor, dignity and the framing of multiculturalist values. In: Kretzmer D, Klein E (eds) Concept of human dignity in human rights discourse. Columbia University Press, pp 263–294
Winfield AF, Jirotka M (2018) Ethical governance is essential to building trust in robotics and artificial intelligence systems. Philos Transact R Soc A Math Phys Eng Sci 376(2133):20180085
Zardiashvili L, Fosch-Villaronga E (2020) "Oh, Dignity too?" Said the Robot: human dignity as the basis for the governance of robotics. Minds and machines. Springer, pp 30:121–143.